

BulPriSA Science Speed-Round Reviewer

Junior High School Science, Grades 7–10 | SY 2026–2027

Based on the DepEd MATATAG Three-Term Budgets of Work (Science 7–10), with a K-12 MELC appendix for Grade 10

How to read this reviewer

Format assumed: multiple choice (conceptual and applied), about 20 s per question, all grades mixed.

Card line: **Q:** cue → **A:** *answer* with a difficulty tag: ^E easy, ^A average, ^D difficult.

Orange boxes list common confusions. These are the pairs most likely to show up as *distractors* in multiple choice, so learn the distinguisher, not just the term.

Items tagged ^{VERIFY} were uncertain or conflicting across sources. They are collected in *Items to verify* at the end.

Drill order: grade sections first (cover the answer, say it aloud), then *Rapid Fire*, then *Cheat Sheets* the night before.

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1 Grade 7

1.1 Term 1: Matter, Investigations, Solutions, Cells

Models and the Particle Model of Matter

KEY FACTS

- A **model** is a simplified representation of something too small, large, or hidden to observe directly.
- Particle model:** all matter is made of tiny particles, and each pure substance has its own kind.
- Particles are always moving, have spaces between them, attract each other, and move faster when heated.
- Solid:** packed, fixed positions, vibrate only, so it has definite shape and volume.
- Liquid:** close but slide past each other, so it has definite volume and takes the container's shape.
- Gas:** far apart and moving fast and randomly, so it has no fixed shape or volume and is compressible.
- Heat *absorbed*: melting, evaporation/boiling, sublimation. Heat *released*: freezing, condensation, deposition.
- A pure substance stays at constant temperature while it changes state.

QUICK CARDS

- Q: Simplified representation of something hard to observe → **A:** *Model*^E
- Q: State with fixed shape and fixed volume → **A:** *Solid*^E
- Q: State most easily compressed → **A:** *Gas*^E
- Q: Particles only vibrate in fixed positions → **A:** *Solid*^A
- Q: Solid changes directly to gas → **A:** *Sublimation*^E
- Q: Gas changes directly to solid → **A:** *Deposition*^A
- Q: Mothballs shrinking in a closet → **A:** *Sublimation*^A
- Q: Droplets forming outside a glass of iced water → **A:** *Condensation*^E
- Q: Effect of heating on particle motion → **A:** *Particles move faster*^E
- Q: Food colouring spreads faster in hot or cold water? → **A:** *Hot water*^A
- Q: Temperature of pure ice while it is melting → **A:** *Stays constant (0 °C)*^D
- Q: Why gases can be compressed → **A:** *Large spaces between particles*^A
- Q: Changes of state that release heat → **A:** *Freezing, condensation, deposition*^D

Common confusions

Evaporation vs. **boiling**: evaporation happens only at the surface and at any temperature; boiling happens throughout the liquid at the boiling point.

Melting vs. **dissolving**: melting is solid → liquid by heating one substance; dissolving is a solute spreading through a solvent.

Sublimation vs. **deposition**: sublimation is solid → gas; deposition is gas → solid.

Scientific Investigations and Measurement

KEY FACTS

- Report parts (BOW): aim/problem → materials and equipment → method → results (data) → conclusion.
- Independent** variable is what you change; **dependent** is what you measure; **controlled** variables are kept the same.
- SI base units: m, kg, s, K, mol, A, cd (see Cheat Sheets).
- Read liquid volume at the *bottom of the meniscus*, with your eye level to it.
- Repeat trials to make results more reliable.

QUICK CARDS

- Q: Variable deliberately changed → **A:** *Independent variable*^E
- Q: Variable measured as the result → **A:** *Dependent variable*^E
- Q: Factors kept the same in a fair test → **A:** *Controlled variables*^E
- Q: SI base unit of mass → **A:** *Kilogram*^E
- Q: SI base unit of temperature → **A:** *Kelvin*^A
- Q: SI base unit of amount of substance → **A:** *Mole*^D
- Q: Tool for measuring liquid volume → **A:** *Graduated cylinder*^E
- Q: Where to read a water meniscus → **A:** *Bottom of the curve, at eye level*^A
- Q: 1 L equals → **A:** *1000 mL*^E
- Q: Prefix meaning one-thousandth → **A:** *milli-*^A
- Q: Part of a report that answers the aim → **A:** *Conclusion*^E
- Q: Testing how water temperature affects sugar's dissolving time: dependent variable? → **A:** *Time to dissolve*^D
- Q: Repeating trials improves the → **A:** *Reliability*^A

Common confusions

Accuracy vs. **precision**: accuracy means close to the true value; precision means repeated readings are close to each other.

Reliability vs. **validity**: reliable results repeat consistently; a valid test actually measures what it claims to.

Solutions, Solubility, Acids and Bases

KEY FACTS

- A **solution** is a homogeneous mixture of a **solute** (dissolved, usually the smaller amount) in a **solvent** (usually the larger amount).
- Water is the “universal solvent.”
- Concentration: % by mass = $\frac{\text{mass of solute}}{\text{mass of solution}} \times 100$.
- Unsaturated** solutions can dissolve more solute; **saturated** ones cannot; **supersaturated** ones hold more than normal (unstable).
- Solubility depends on temperature, the nature of solute and solvent (“like dissolves like”), and pressure (for gases).
- Most solids dissolve better when hot, but gases dissolve *less* when warm.
- Stirring and crushing speed up *dissolving*; they do not change solubility.
- Litmus: acid turns blue litmus red; base turns red litmus blue; neutral salt solutions cause no change.

QUICK CARDS

- Q: Component present in the greater amount → **A:** *Solvent*^E
- Q: Universal solvent → **A:** *Water*^E
- Q: Solution that can still dissolve more solute → **A:** *Unsaturated*^E
- Q: Holds more solute than normally possible → **A:** *Supersaturated*^A
- Q: Blue litmus dipped in vinegar turns → **A:** *Red*^E
- Q: Red litmus dipped in soap solution turns → **A:** *Blue*^E
- Q: pH of pure water at 25 °C → **A:** *7 (neutral)*^A
- Q: Why a warm soft drink goes flat faster → **A:** *Gas solubility falls as temperature rises*^D
- Q: Stirring changes solubility or rate of dissolving? → **A:** *Rate of dissolving only*^D
- Q: 20 g sugar in 80 g water: % by mass → **A:** *20%*^D
- Q: Air is a gaseous solution; its solvent → **A:** *Nitrogen*^D
- Q: Brass is a solid solution of → **A:** *Copper and zinc*^A
- Q: Does oil dissolve in water? → **A:** *No (“like dissolves like”)*^A

Common confusions

Solubility vs. **rate of dissolving**: solubility is *how much* dissolves; rate is *how fast*.

Concentrated vs. **saturated**: concentrated means a lot of solute; saturated means no more can dissolve at that temperature.

Mnemonic: **BRA**: Blue → Red = Acid (so red → blue = base).

The Compound Microscope**KEY FACTS**

- Total magnification = eyepiece × objective (e.g., $10 \times 40 \times = 400 \times$).
- Objectives: scanner ($4 \times$), LPO ($10 \times$), HPO ($40 \times$), and oil immersion ($100 \times$, needs oil).
- **Diaphragm** controls light; **stage clips** hold the slide; **nose-piece** rotates the objectives.
- Coarse knob is for the lowest power only; use only the **fine** knob on HPO.
- Carry with one hand on the arm and one under the base. Start with the lowest-power objective.
- The image is inverted (upside down and reversed). Moving to higher power gives a smaller, dimmer field of view.

QUICK CARDS

Q: Magnification with $10 \times$ eyepiece and $40 \times$ objective → **A**: $400 \times$ ^E

Q: Knob to use with the HPO → **A**: *Fine adjustment* ^A

Q: Part that regulates light → **A**: *Diaphragm* ^E

Q: Holds the slide in place → **A**: *Stage clips* ^E

Q: Rotating part holding the objectives → **A**: *Revolving nose-piece* ^E

Q: Lens nearest the eye → **A**: *Eyepiece (ocular)* ^E

Q: Letter “e” as seen through the microscope → **A**: *Upside down and reversed* ^A

Q: Correct way to carry it → **A**: *One hand on arm, one under base* ^E

Q: Objective to focus with first → **A**: *Lowest power (scanner)* ^A

Q: LPO to HPO: field of view becomes → **A**: *Smaller and darker* ^D

Q: The $100 \times$ objective requires → **A**: *Immersion oil* ^D

Common confusions

Coarse vs. **fine adjustment**: coarse moves the stage a lot (low power only); fine makes small focus changes.

Magnification vs. **resolution**: magnification enlarges the image; resolution separates two close points clearly.

Plant and Animal Cells**KEY FACTS**

- The cell is the basic unit of life. **Robert Hooke** named “cells” from cork (1665).
- Cell theory: **Schleiden** (plants), **Schwann** (animals), **Virchow** (cells come from pre-existing cells).
- **Nucleus** is the control centre (DNA); **cell membrane** controls entry and exit; **cytoplasm** is the jelly-like interior.
- **Mitochondria** carry out respiration (the “powerhouse”); **chloroplasts** carry out photosynthesis; **ribosomes** make proteins.
- Plant cells only: cell wall (cellulose), chloroplasts, and a large central vacuole. Animal cells are irregular and have small vacuoles.
- Organisms are **unicellular** (bacteria, amoeba) or **multicellular** (humans).

QUICK CARDS

Q: Who coined the term “cell” → **A**: *Robert Hooke* ^E

Q: “All cells come from pre-existing cells” → **A**: *Rudolf Virchow* ^A

Q: Proposed that all plants are made of cells → **A**: *Matthias Schleiden* ^A

Q: Powerhouse of the cell → **A**: *Mitochondrion* ^E

Q: Organelle for photosynthesis → **A**: *Chloroplast* ^E

Q: Control centre of the cell → **A**: *Nucleus* ^E

Q: Material of the plant cell wall → **A**: *Cellulose* ^A

Q: Organelle that makes proteins → **A**: *Ribosome* ^A

Q: Example of a unicellular organism → **A**: *Bacterium / amoeba* ^E

Q: Cell without a true nucleus → **A**: *Prokaryotic cell* ^D

Q: Plant organelle that keeps the cell firm (turgid) → **A**: *Large central vacuole* ^D

Q: Organelle absent in onion bulb epidermis cells → **A**: *Chloroplast* ^D

Q: Common stain for cheek cells → **A**: *Methylene blue* ^A

Common confusions

Cell wall vs. **cell membrane**: the wall is a rigid outer layer (plants only); the membrane is a flexible, selective barrier (all cells).

Chloroplast vs. **chlorophyll**: the chloroplast is the organelle; chlorophyll is the green pigment inside it.

Prokaryote vs. **eukaryote**: a prokaryote has no nucleus (bacteria); a eukaryote has a nucleus (plants, animals, fungi).

1.2 Term 2: Reproduction, Ecology, Forces, Motion**Cellular Reproduction****KEY FACTS**

- **Mitosis** gives 2 identical cells and is used for growth and repair in body cells.
- **Meiosis** gives 4 genetically different sex cells (gametes) with half the chromosomes.
- Human body cell: 46 chromosomes; gamete: 23.
- Stages: interphase (DNA copies), then prophase, metaphase, anaphase, telophase, then cytokinesis.
- Fertilization: sperm + egg → zygote.
- **Sexual** reproduction: 2 parents, varied offspring. **Asexual**: 1 parent, identical offspring.
- Asexual types: binary fission (bacteria), budding (yeast, hydra), spores (ferns, fungi), vegetative propagation (cuttings), fragmentation.

QUICK CARDS

Q: Cell division for growth and repair → **A**: *Mitosis* ^E

Q: Cell division that makes gametes → **A**: *Meiosis* ^E

Q: Daughter cells from one meiosis → **A**: *Four* ^E

Q: Chromosomes in a human sperm cell → **A**: *23* ^A

Q: Cell formed when sperm and egg fuse → **A**: *Zygote* ^E

Q: Chromosomes line up at the middle → **A**: *Metaphase* ^A

Q: Sister chromatids pulled apart → **A**: *Anaphase* ^A

Q: Stage when DNA is copied → **A**: *Interphase* ^D

Q: Division of the cytoplasm → **A**: *Cytokinesis* ^A

Q: Yeast reproduces asexually by → **A**: *Budding* ^A

Q: Bacteria reproduce by → **A**: *Binary fission* ^A

Q: Offspring genetically identical to the parent → **A**: *Asexual reproduction* ^E

Q: Ferns and mushrooms reproduce asexually by → **A**: *Spores* ^A

Q: Why siblings are not identical → **A**: *Meiosis + random fertilization* ^D

Common confusions

Mitosis vs. **meiosis**: mitosis: 1 division, 2 identical body cells. Meiosis: 2 divisions, 4 varied gametes.

Diploid vs. **haploid**: diploid cells have 2 sets (46 in humans); haploid gametes have 1 set (23).

Chromosome vs. **chromatid**: a chromatid is one of the two identical copies of a replicated chromosome, joined at the centromere.

Mnemonic: IPMAT: Interphase, Prophase, Metaphase (Middle), Anaphase (A)part, Telophase (T)wo nuclei).

Levels of Biological Organization**KEY FACTS**

- Cell → tissue → organ → organ system → organism → population → community → ecosystem → (biome) → biosphere.
- Population**: one species in one area. **Community**: all the populations that interact there.
- Ecosystem** = community + nonliving (abiotic) factors. **Biosphere**: all life on Earth.

QUICK CARDS

Q: Group of similar cells doing one job → A: *Tissue*^E

Q: Same species living in one area → A: *Population*^E

Q: Community plus nonliving factors → A: *Ecosystem*^E

Q: Highest level of organization → A: *Biosphere*^E

Q: The heart is an example of → A: *An organ*^E

Q: Level right after organ → A: *Organ system*^E

Q: All tamaraws in Mindoro → A: *Population*^A

Q: Fish, algae, corals, water, and sunlight in a reef → A: *Ecosystem*^A

Q: Blood is classified as a → A: *Tissue (connective)*^D

Common confusions

Population vs. **community**: a population is one species; a community is many species living together.

Community vs. **ecosystem**: an ecosystem adds the abiotic factors (water, soil, light).

Trophic Levels and Energy Transfer**KEY FACTS**

- The Sun is the ultimate energy source. Producers (autotrophs) form the 1st trophic level.
- Primary consumers are herbivores; secondary consumers eat herbivores; then tertiary consumers.
- Decomposers (bacteria, fungi) recycle dead matter.
- 10% rule**: only about 10% of energy passes to the next level. The rest is used or lost as heat.
- Energy loss limits food chains to about 4–5 levels. A pyramid of energy is always upright.

QUICK CARDS

Q: Organisms at the 1st trophic level → A: *Producers*^E

Q: Grass → grasshopper → frog → snake: secondary consumer → A: *Frog*^A

Q: Energy passed to the next level → A: *About 10%*^E

Q: Form of most energy “lost” between levels → A: *Heat*^A

Q: Organisms that break down dead matter → A: *Decomposers*^E

Q: Another name for primary consumers → A: *Herbivores*^E

Q: Interconnected food chains → A: *Food web*^E

Q: Producers hold 10 000 kJ; primary consumers get about → A: 1000 kJ^D

Q: Why food chains rarely exceed 5 levels → A: *Too little energy left*^D

Q: Scientist linked to the 10% rule → A: *Raymond Lindeman*^D

Common confusions

Food chain vs. **food web**: a chain is one feeding path; a web is many interlinked chains.

Pyramid of energy vs. **pyramid of numbers**: the energy pyramid is always upright; a numbers pyramid can be inverted (one tree, many insects).

Balanced and Unbalanced Forces**KEY FACTS**

- A force is a push or pull, measured in **newtons (N)** with a spring balance.
- Balanced** forces (net force 0) mean the object is at rest or moving at constant velocity.
- Unbalanced** forces change speed or direction (acceleration).
- A free-body diagram uses arrows whose length shows size and whose head shows direction.
- Weight $W = mg$, with $g \approx 9.8 \text{ m/s}^2$.

QUICK CARDS

Q: SI unit of force → A: *Newton*^E

Q: Device for measuring force → A: *Spring balance*^E

Q: Net force on a car at constant velocity → A: *Zero*^A

Q: Freely falling mango: balanced or unbalanced? → A: *Unbalanced*^E

Q: Box resting on a ramp: balanced or unbalanced? → A: *Balanced*^A

Q: Diagram of all forces on one object → A: *Free-body diagram*^E

Q: Force that opposes motion between surfaces → A: *Friction*^E

Q: Support force perpendicular to a surface → A: *Normal force*^A

Q: 10 N right and 4 N left: net force → A: *6 N to the right*^A

Q: Unbalanced forces change an object's → A: *Speed or direction*^A

Q: Weight of a 2 kg object ($g = 9.8$) → A: *19.6 N*^D

Common confusions

Mass vs. **weight**: mass is the amount of matter (kg, same everywhere); weight is the pull of gravity (N, changes with g).

Balanced forces vs. **no forces**: a book at rest still has forces acting on it; they just cancel.

Motion: Distance, Displacement, Speed, Velocity**KEY FACTS**

- Position is described from a **reference point**.
- Distance** is total path length (scalar). **Displacement** is the straight-line change in position with direction (vector).
- Speed $v = d/t$ (scalar). Velocity = displacement/time (vector).
- On a distance–time graph, a straight sloped line means uniform velocity, the slope is the speed, and a flat line means at rest.
- $1 \text{ m/s} = 3.6 \text{ km/h}$.

QUICK CARDS

Q: Quantity with magnitude and direction → A: *Vector*^E

Q: Is speed scalar or vector? → A: *Scalar*^E

Q: Point used to describe position → A: *Reference point*^E

Q: 3 m east then 3 m west: distance, displacement → A: *6 m; 0 m*^A

Q: Car covers 100 m in 5 s: speed → A: *20 m/s*^E

Q: Slope of a distance–time graph → A: *Speed*^A

Q: Flat (horizontal) line on a d–t graph → A: *At rest*^A

Q: Straight sloped line on a d–t graph → A: *Uniform velocity*^A

Q: Steeper d–t line means → A: *Faster*^A

Q: 72 km/h in m/s → A: *20 m/s*^D

Common confusions

Distance vs. displacement: distance is the whole path; displacement is start to finish with direction.

Speed vs. velocity: velocity includes direction, so a turning car at constant speed has changing velocity.

1.3 Term 3: Heat, Earthquakes, the Sun and Weather

Heat Transfer

KEY FACTS

- Heat** is energy transferred because of a temperature difference (joules). **Temperature** measures the average kinetic energy of particles ($^{\circ}\text{C}$, K).
- Heat flows from hot to cold.
- Conduction** is transfer by particle collisions in contact (best in metals).
- Convection** is transfer by moving fluids: warm fluid is less dense and rises.
- Radiation** is transfer by electromagnetic waves and needs no medium (Sun to Earth).
- $K = ^{\circ}\text{C} + 273$.
- Heat-to-electricity devices: thermoelectric generators and geothermal power plants (e.g., Tiwi, Albay).

QUICK CARDS

- Q: Metal spoon heating up in hot soup → A: *Conduction*^E
- Q: How the Sun's heat reaches Earth → A: *Radiation*^E
- Q: Circulating currents in boiling water → A: *Convection*^E
- Q: Transfer that works in a vacuum → A: *Radiation*^A
- Q: Why pot handles are wood or plastic → A: *They are insulators*^E
- Q: SI unit of heat → A: *Joule*^E
- Q: Measure of average particle kinetic energy → A: *Temperature*^A
- Q: 25°C in kelvin → A: 298 K ^A
- Q: Why warm fluids rise → A: *Less dense*^A
- Q: Dark shirts get hotter in sunlight because they → A: *Absorb more radiation*^A
- Q: Why air conditioners are mounted high → A: *Cool air sinks (convection)*^D
- Q: Device turning a temperature difference directly into electricity → A: *Thermoelectric generator*^D
- Q: Geothermal plant in Albay → A: *Tiwi*^D

Common confusions

Heat vs. temperature: heat is energy in transit (J); temperature is how hot ($^{\circ}\text{C}$, K).

Conduction vs. convection: conduction is particles passing energy without flowing; convection is the fluid itself moving.

Faults, Earthquakes, and Tsunamis

KEY FACTS

- A **fault** is a crack in the crust along which blocks have moved. The block above an inclined fault is the hanging wall; the block below is the footwall.
- Normal fault:** hanging wall moves down (tension). **Reverse/thrust:** hanging wall moves up (compression). **Strike-slip:** horizontal sliding (shear).
- Focus** (hypocenter) is where the rupture starts underground; the **epicenter** is directly above it on the surface.
- Magnitude** is the energy released (one number per quake). **Intensity** is the shaking and damage at a place; PHIVOLCS uses PEIS I–X.

- The Philippine Fault Zone is a left-lateral strike-slip fault about 1200 km long. The West Valley Fault threatens Metro Manila.
- Find the nearest active fault with **PHIVOLCS Fault-Finder**.
- Tsunamis are mostly caused by undersea quakes that move the seafloor vertically. A natural warning is the sea suddenly pulling back.
- During shaking: **Duck, Cover, and Hold**.

QUICK CARDS

- Q: Point on the surface above the focus → A: *Epicenter*^E
- Q: Underground origin of a quake → A: *Focus (hypocenter)*^E
- Q: Fault where the hanging wall drops → A: *Normal fault*^E
- Q: Fault where the hanging wall rises → A: *Reverse fault*^E
- Q: Blocks slide horizontally → A: *Strike-slip fault*^E
- Q: Stress that forms normal faults → A: *Tension*^A
- Q: Stress that forms reverse faults → A: *Compression*^A
- Q: Movement type of the Philippine Fault Zone → A: *Left-lateral strike-slip*^D
- Q: Approximate length of the Philippine Fault Zone → A: *About 1200 km*^D
- Q: PHIVOLCS intensity scale → A: *PEIS (I to X)*^A
- Q: Magnitude measures → A: *Energy released*^A
- Q: Instrument recording ground shaking → A: *Seismograph*^E
- Q: PH agency for quakes and volcanoes → A: *PHIVOLCS*^E
- Q: What to do during shaking → A: *Duck, Cover, and Hold*^E
- Q: Natural sign of an incoming tsunami → A: *Sea suddenly recedes*^A
- Q: Fault system threatening Metro Manila → A: *West Valley Fault*^A
- Q: PHIVOLCS tool to find the nearest fault → A: *FaultFinder*^A
- Q: Fault type most likely to cause a tsunami → A: *Reverse/thrust (vertical seafloor shift)*^D
- Q: Magnitude of the 2013 Bohol earthquake → A: 7.2 ^D
- Q: Fault types most common in PH, and why → A: *Reverse/thrust and strike-slip, from plate convergence (expected DepEd answer)*^{[VERIFY] D}

Common confusions

Magnitude vs. intensity: magnitude is one value for the whole quake; intensity differs from place to place.

Focus vs. epicenter: the focus is underground; the epicenter is on the surface above it.

Tsunami vs. storm surge: a tsunami comes from seafloor displacement (quake or eruption); a storm surge is seawater pushed by typhoon winds and low pressure.

The Sun's Influence: Seasons, Breezes, Monsoons, ITCZ

KEY FACTS

- Earth's axis is tilted about 23.5° . Rotation (about 24 h) causes day and night. Revolution (about $365\frac{1}{4}$ days) plus the tilt causes the seasons.
- Direct rays are more intense; slanting rays spread over a larger area.
- The PH is in the Northern Hemisphere: longest day at the June solstice, shortest at the December solstice. Equinoxes (March and September) have about equal day and night.
- Land heats and cools faster than water. Daytime **sea breeze** blows sea → land; nighttime **land breeze** blows land → sea.
- Amihan** (northeast monsoon) is cool and dry. **Habagat** (southwest monsoon) is warm, moist, and rainy.
- ITCZ:** a low-pressure belt near the equator where the trade winds meet, bringing clouds and rain.
- The ozone layer absorbs most UV radiation.

QUICK CARDS

Q: Main cause of the seasons → A: *Tilt of Earth's axis*^E

Q: Tilt of Earth's axis → A: *About 23.5°*^E

Q: Day and night are caused by → A: *Rotation*^E

Q: Time for one revolution → A: *About 365 $\frac{1}{4}$ days*^E

Q: Daytime breeze from sea to land → A: *Sea breeze*^E

Q: Why land heats faster than the sea → A: *Water has a higher heat capacity*^A

Q: Local name of the northeast monsoon → A: *Amihan*^E

Q: Local name of the southwest monsoon → A: *Habagat*^E

Q: Monsoon behind heavy PH rains in the wet season → A: *Habagat*^A

Q: Belt where the trade winds converge → A: *ITCZ*^A

Q: Longest daytime in the PH → A: *June solstice*^A

Q: About equal day and night → A: *Equinox*^E

Q: Gas layer that absorbs UV → A: *Ozone*^A

Q: Why slanting rays heat less → A: *Energy spread over a larger area*^D

Q: Earth is closest to the Sun in January, yet it is the cool season in the PH. This shows → A: *Seasons aren't caused by distance*^D

Common confusions

Rotation vs. **revolution**: rotation is spinning on the axis (day/night); revolution is orbiting the Sun (year, seasons).

Sea breeze vs. **land breeze**: a breeze is named after where it comes *from*: sea breeze by day, land breeze by night.

Amihan vs. **Habagat**: Amihan is NE, cool and dry, about Nov–Feb; Habagat is SW, rainy, about Jun–Sep.

Weather vs. **climate**: weather is short-term conditions; climate is the long-term (about 30-year) average.

Cards in this part: 165 (E 75, A 59, D 31)

2 Grade 8

2.1 Term 1: Body Systems, Heredity, Taxonomy, Energy in Life, Atoms

Organ Systems Working Together

KEY FACTS

- Food path: mouth → pharynx → esophagus → stomach → small intestine → large intestine → rectum → anus.
- Processes: ingestion, mechanical processing (chewing, churning), secretion, chemical digestion (enzymes), absorption, elimination.
- Peristalsis**: wave-like muscle contractions that push food along.
- Saliva has amylase (digests starch). The stomach adds HCl and pepsin (protein). The liver makes bile, which the gallbladder stores; bile emulsifies fats.
- The small intestine (duodenum, jejunum, ileum) absorbs nutrients through its **villi**. The large intestine absorbs water.
- Excretion: kidneys (nephrons) filter blood into urine. The lungs remove CO₂; the skin removes sweat.
- Plant transport: roots absorb water, **xylem** carries water up, **phloem** carries sugars, and transpiration at the stomata pulls water up.

QUICK CARDS

- Q: Wave-like contractions that move food → **A: Peristalsis**^E
- Q: Where most digestion and absorption happen → **A: Small intestine**^E
- Q: Finger-like projections that absorb nutrients → **A: Villi**^A
- Q: Starch-digesting enzyme in saliva → **A: Salivary amylase**^A
- Q: Organ that produces bile → **A: Liver**^E
- Q: Organ that stores bile → **A: Gallbladder**^A
- Q: Role of bile → **A: Emulsifies fats**^D
- Q: Protein-digesting enzyme of the stomach → **A: Pepsin**^A
- Q: Main job of the large intestine → **A: Absorbs water**^A
- Q: Tube from the pharynx to the stomach → **A: Esophagus**^E
- Q: Flap that keeps food out of the windpipe → **A: Epiglottis**^A
- Q: First part of the small intestine → **A: Duodenum**^D
- Q: Functional unit of the kidney → **A: Nephron**^D
- Q: Plant tissue that carries water upward → **A: Xylem**^E
- Q: Plant tissue that carries sugars → **A: Phloem**^E
- Q: Water loss through leaf stomata → **A: Transpiration**^A

Common confusions

Mechanical vs. **chemical digestion**: mechanical breaks food into smaller pieces; chemical uses enzymes to break molecules apart.

Digestion vs. **absorption**: digestion breaks food down; absorption moves nutrients into the blood.

Excretion vs. **elimination**: excretion removes metabolic waste (urine, CO₂); elimination (egestion) removes undigested food (feces).

Xylem vs. **phloem**: xylem carries water up only; phloem carries food both ways.

Mnemonic: Phloem carries food (same “f” sound); Xylem carries water up to the sky.

Heredity: Patterns of Inheritance

KEY FACTS

- Gregor Mendel**, the “father of genetics,” worked with garden peas.
- Alleles are versions of a gene. The dominant allele is written in capitals (T); the recessive in lowercase (t).

- Genotype** is the allele letters (TT, Tt, tt); **phenotype** is the visible trait.
- Homozygous: TT or tt. Heterozygous: Tt.
- Tt × Tt gives genotypes 1 TT : 2 Tt : 1 tt and phenotypes 3 dominant : 1 recessive.
- Tt × tt gives 1 : 1.
- Laws: Dominance, Segregation, Independent Assortment. A Punnett square predicts the offspring.
- Pedigree symbols: square = male, circle = female, shaded = affected.

QUICK CARDS

- Q: Father of genetics → **A: Gregor Mendel**^E
- Q: Mendel’s experimental plant → **A: Garden pea**^E
- Q: Observable trait → **A: Phenotype**^E
- Q: Allele combination (letters) → **A: Genotype**^E
- Q: Tt is described as → **A: Heterozygous**^E
- Q: Alternative forms of a gene → **A: Alleles**^A
- Q: Grid used to predict offspring → **A: Punnett square**^E
- Q: Phenotype ratio of Tt × Tt → **A: 3:1**^A
- Q: Genotype ratio of Tt × Tt → **A: 1:2:1**^A
- Q: Family chart of a trait across generations → **A: Pedigree**^A
- Q: Shaded square in a pedigree → **A: Affected male**^A
- Q: Tt × tt: chance of a short (tt) offspring → **A: 50%**^D
- Q: tt × tt: chance of a tall offspring → **A: 0%**^D
- Q: Law: allele pairs separate when gametes form → **A: Law of Segregation**^D

Common confusions

Genotype vs. **phenotype**: genotype is the letters (Tt); phenotype is what you see (tall).

Homozygous vs. **heterozygous**: homozygous alleles are the same (TT or tt); heterozygous alleles differ (Tt).

Dominant vs. **common**: dominant means it shows in Tt. It does *not* mean the trait is most frequent.

Taxonomic Classification

KEY FACTS

- Carolus Linnaeus** is the father of taxonomy. He introduced binomial nomenclature, written *Genus species* (italicized, genus capitalized).
- Ranks: Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species.
- Three domains (**Carl Woese**): Bacteria, Archaea, Eukarya.
- Six kingdoms: Eubacteria, Archaeobacteria, Protista, Fungi, Plantae, Animalia.
- Humans: Eukarya, Animalia, Chordata, **Mammalia**, **Primates**, Hominidae, *Homo sapiens*.
- Mammals have hair and mammary glands and are warm-blooded. Primates have opposable thumbs, forward-facing eyes, nails, and large brains.

QUICK CARDS

- Q: Father of taxonomy → **A: Carolus Linnaeus**^E
- Q: Two-name naming system → **A: Binomial nomenclature**^E
- Q: Broadest taxon → **A: Domain**^E
- Q: Most specific taxon → **A: Species**^E
- Q: Taxon between order and genus → **A: Family**^A
- Q: Proposed the three-domain system → **A: Carl Woese**^D
- Q: The three domains → **A: Bacteria, Archaea, Eukarya**^A
- Q: Kingdom of mushrooms and molds → **A: Fungi**^E
- Q: Kingdom of amoebas and most algae → **A: Protista**^A
- Q: Kingdom of prokaryotes living in extreme conditions → **A: Archaeobacteria**^D

- Q: Class of humans → A: *Mammalia*^E
 Q: Order of humans → A: *Primates*^E
 Q: Scientific name of humans → A: *Homo sapiens*^E
 Q: Two features unique to mammals → A: *Hair and mammary glands*^A
 Q: Primate trait for grasping → A: *Opposable thumb*^A
 Q: Scientific name of the Philippine eagle → A: *Pithecophaga jefferyi*^D

Common confusions

Domain vs. kingdom: domain is the rank *above* kingdom (3 domains, 6 kingdoms).

Archaea vs. Bacteria: both are prokaryotes, but archaea differ chemically and often live in extreme environments.

Mnemonic: Dear King Philip Came Over For Good Soup.

Photosynthesis, Respiration, and Cycles in Nature

KEY FACTS

- Photosynthesis (in the chloroplast): $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow[\text{chlorophyll}]{\text{light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
- Cellular respiration (in the mitochondrion): $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy (ATP)}$.
- Stomata exchange gases, and guard cells open and close them.
- Starch test: iodine turns blue-black. Destarch the plant in the dark before testing.
- Anaerobic respiration: yeast makes ethanol and CO_2 ; muscles make lactic acid.
- Carbon cycle: photosynthesis removes CO_2 ; respiration, combustion, and decay return it.
- Water cycle: evaporation, transpiration, condensation, precipitation, runoff/collection.

QUICK CARDS

- Q: Gas released by photosynthesis → A: *Oxygen*^E
 Q: Raw materials of photosynthesis → A: *CO₂, water, light*^E
 Q: Green pigment that absorbs light → A: *Chlorophyll*^E
 Q: Leaf pores for gas exchange → A: *Stomata*^E
 Q: Cells that open and close the stomata → A: *Guard cells*^A
 Q: Formula of glucose → A: *C₆H₁₂O₆*^A
 Q: Energy molecule made in respiration → A: *ATP*^A
 Q: Reagent to test a leaf for starch → A: *Iodine solution*^A
 Q: Colour of iodine with starch → A: *Blue-black*^A
 Q: Why a plant is kept in the dark before a starch test → A: *To use up stored starch*^D
 Q: Products of yeast fermentation → A: *Ethanol and CO₂*^D
 Q: Muscles short of O₂ produce → A: *Lactic acid*^D
 Q: Light-independent stage in the stroma → A: *Calvin cycle*^D
 Q: Water falling from clouds → A: *Precipitation*^E
 Q: Gas added by burning fossil fuels → A: *CO₂*^E

Common confusions

Photosynthesis vs. respiration: photosynthesis stores energy (uses CO_2 , makes O_2); respiration releases it (uses O_2 , makes CO_2). Plants do *both*.

Aerobic vs. anaerobic: aerobic uses oxygen and yields much ATP; anaerobic has no oxygen and yields little ATP.

Evaporation vs. transpiration: transpiration is evaporation of water specifically from plant leaves.

Development of the Atomic Model

KEY FACTS

- Democritus:** “atomos” (indivisible).
- Dalton** (early 1800s): solid-sphere (billiard-ball) model and atomic theory.

- J.J. Thomson** (1897): discovered the electron with a cathode-ray tube; plum-pudding model.
- Rutherford** (1911): gold-foil experiment showed a small, dense, positive **nucleus** and mostly empty space.
- Bohr** (1913): electrons in fixed energy levels (shells); planetary model.
- Modern view: electron-cloud (quantum mechanical) model, **Schrödinger**.
- Chadwick** (1932) discovered the neutron.
- Shells for the first 20 elements fill 2, 8, 8, 2.

QUICK CARDS

- Q: Coined “atomos” → A: *Democritus*^A
 Q: Solid-sphere (billiard-ball) model → A: *John Dalton*^E
 Q: Plum-pudding model → A: *J.J. Thomson*^E
 Q: Particle found with the cathode-ray tube → A: *Electron*^A
 Q: Gold-foil experiment → A: *Ernest Rutherford*^E
 Q: Main conclusion of the gold-foil experiment → A: *Dense positive nucleus; mostly empty space*^A
 Q: Electrons in fixed energy levels → A: *Niels Bohr*^E
 Q: Discovered the neutron → A: *James Chadwick*^A
 Q: Current model of the atom → A: *Electron-cloud (quantum) model*^A
 Q: Maximum electrons in the 2nd shell → A: *8*^A
 Q: Electron arrangement of sodium ($Z = 11$) → A: *2, 8, 1*^D

Common confusions

Thomson vs. Rutherford model: Thomson: positive “pudding” with electrons, no nucleus. Rutherford: a tiny nucleus with electrons around it.

Bohr vs. electron-cloud model: Bohr places electrons in exact orbits; the cloud model gives only regions of probability.

Subatomic Particles

KEY FACTS

- Proton** (p^+): charge +1, mass about 1 amu, in the nucleus.
- Neutron** (n^0): no charge, mass about 1 amu, in the nucleus.
- Electron** (e^-): charge -1, mass about 1/1836 of a proton, in the shells.
- Atomic number Z = number of protons. Mass number A = protons + neutrons. In a neutral atom, electrons = protons.

QUICK CARDS

- Q: Charge of a neutron → A: *Zero (neutral)*^E
 Q: Particle with the smallest mass → A: *Electron*^E
 Q: Atomic number equals the number of → A: *Protons*^E
 Q: Mass number equals → A: *Protons + neutrons*^E
 Q: Where protons and neutrons are found → A: *Nucleus*^E
 Q: About how many times heavier a proton is than an electron → A: *About 1836*^D

Common confusions

Atomic number vs. mass number: atomic number counts protons only; mass number counts protons + neutrons.

Mass number vs. atomic mass: mass number is a whole count for one isotope; atomic mass is the decimal average shown on the table.

Mnemonic: P-E-N: Proton Positive, Electron nEgative, Neutron Neutral.

2.2 Term 2: Elements, Periodic Table, Earth's Crust and Volcanoes

Elements and Compounds

KEY FACTS

- Pure substances have a fixed composition: **elements** and **compounds**.
- Every atom of an element has the same, unique number of protons.
- Compounds are two or more elements chemically combined in a fixed ratio, separable only by chemical means (e.g., electrolysis).
- Mixtures have a variable composition and are homogeneous (solutions) or heterogeneous.

QUICK CARDS

- Q: Substance with only one kind of atom → A: *Element*^E
- Q: Water: element, compound, or mixture? → A: *Compound*^E
- Q: What makes each element unique → A: *Its number of protons*^A
- Q: Is air a pure substance? → A: *No, it is a mixture*^A
- Q: Process that splits water into H₂ and O₂ → A: *Electrolysis*^A
- Q: Ratio of H to O atoms in water → A: *2:1*^E
- Q: Seawater is a → A: *Homogeneous mixture*^A
- Q: Can a compound be separated by filtration? → A: *No, only by chemical means*^D

Common confusions

Compound vs. **mixture**: a compound has a fixed ratio and new properties; a mixture has a variable ratio and its parts keep their properties.

Atom vs. **molecule**: a molecule is two or more atoms bonded together (O₂, H₂O).

The Periodic Table

KEY FACTS

- **Döbereiner**: triads. **Newlands**: law of octaves. **Mendeleev** (1869): ordered by atomic mass, left gaps, and predicted gallium and germanium. **Moseley**: ordered by atomic number (the modern periodic law).
- The table has 7 periods (rows) and 18 groups (columns).
- Elements in the same group have the same number of valence electrons and similar properties. The period number is the number of occupied shells.
- Group 1 alkali metals, Group 2 alkaline earth metals, Groups 3–12 transition metals, Group 17 halogens, Group 18 noble gases.
- Latin-based symbols: Na (natrium), K (kalium), Fe (ferrum), Cu (cuprum), Ag (argentum), Au (aurum), Pb (plumbum), Hg (hydrargyrum), Sn (stannum).
- Example: ²⁷₁₃Al has 13 protons, 13 electrons, and 27 – 13 = 14 neutrons.

QUICK CARDS

- Q: Law of triads → A: *Johann Döbereiner*^A
- Q: Law of octaves → A: *John Newlands*^A
- Q: Father of the periodic table → A: *Dmitri Mendeleev*^E
- Q: Arranged elements by atomic number → A: *Henry Moseley*^A
- Q: Mendeleev's "eka-aluminium" is now → A: *Gallium*^D
- Q: Number of periods → A: *7*^E
- Q: Number of groups → A: *18*^E
- Q: Group 18 elements → A: *Noble gases*^E
- Q: Group 17 elements → A: *Halogens*^E
- Q: Group 1 elements → A: *Alkali metals*^E
- Q: Symbol of potassium → A: *K*^E
- Q: Symbol of sodium → A: *Na*^E
- Q: Element with symbol Au → A: *Gold*^E
- Q: Element with symbol Hg → A: *Mercury*^E
- Q: Latin name behind Fe → A: *Ferrum*^A
- Q: Neutrons in aluminium-27 → A: *14*^A

- Q: Valence electrons of Group 2 elements → A: *2*^A
- Q: Element in period 3, group 17 → A: *Chlorine*^A
- Q: Element number 20 → A: *Calcium*^A
- Q: Why noble gases are unreactive → A: *Full outer shell*^A
- Q: Metalloid used in computer chips → A: *Silicon*^A

Common confusions

Group vs. **period**: groups are columns (same valence electrons); periods are rows (same number of shells).

Mendeleev vs. **Moseley**: Mendeleev ordered by atomic *mass*; Moseley ordered by atomic *number*.

Metal vs. **metalloid**: metalloids (B, Si, Ge, As, Sb, Te) lie on the "staircase" and have mixed properties.

Earth's Surface and Crust

KEY FACTS

- About 71% of Earth's surface is water and about 29% is land.
- **Oceanic crust** is thin, dense, basaltic, and young. **Continental crust** is thick, less dense, granitic, and old.
- The upper crustal layers are **sial** (silica + alumina, continental) over **sima** (silica + magnesium, oceanic).
- **Lithosphere** = crust + uppermost mantle. It is rigid and broken into plates.
- Volcanoes, earthquake epicenters, and young mountain belts cluster along plate boundaries, such as the Pacific Ring of Fire.

QUICK CARDS

- Q: Percent of Earth's surface covered by water → A: *About 71%*^E
- Q: Thinner, denser type of crust → A: *Oceanic crust*^A
- Q: Main rock of oceanic crust → A: *Basalt*^A
- Q: Main rock of continental crust → A: *Granite*^A
- Q: Crust plus uppermost mantle → A: *Lithosphere*^A
- Q: Silica–alumina-rich continental layer → A: *Sial*^D
- Q: Where volcanoes, quakes, and mountains cluster → A: *Plate boundaries*^A
- Q: Volcano and quake belt around the Pacific → A: *Ring of Fire*^E

Volcanoes

KEY FACTS

- By activity (PHIVOLCS): **active**, **potentially active**, **inactive**. PHIVOLCS lists 24 active PH volcanoes ^[VERIFY].
- **Shield**: broad, gentle slopes, fluid basaltic lava. **Cinder cone**: small and steep, made of pyroclastics. **Composite (strato)**: lava and ash layers, steep and symmetrical (Mayon).
- Silica content controls viscosity: high silica gives thick magma and explosive eruptions; low silica gives runny lava and gentle eruptions.
- Eruption types: Hawaiian, Strombolian, Vulcanian, Plinian (most violent; Pinatubo 1991), and phreatic (steam-driven; common at Taal).
- A **caldera** is a large basin formed when a volcano collapses (Taal Lake).
- PH examples: Mayon (Albay), Taal (Batangas), Pinatubo (Zambales), Kanlaon (Negros), Bulusan (Sorsogon), Hibok-Hibok (Camiguin).

QUICK CARDS

- Q: Volcano type of Mayon → A: *Composite (stratovolcano)*^E
- Q: Broad, gently sloping volcano of fluid lava → A: *Shield volcano*^E
- Q: Small, steep cone of cinders → A: *Cinder cone*^E
- Q: Large basin from a collapsed volcano → A: *Caldera*^A
- Q: Steam-driven eruption → A: *Phreatic*^A

- Q:** Most violent eruption type, with a tall column → **A:** *Plinian*^A
- Q:** Year Pinatubo erupted → **A:** 1991^E
- Q:** Magma property that makes eruptions explosive → **A:** *High viscosity (high silica)*^D
- Q:** Low-silica lava is → **A:** *Fluid (runny)*^A
- Q:** Volcano in a lake in Batangas → **A:** *Taal*^E
- Q:** Active volcano on Negros → **A:** *Kanlaon*^A
- Q:** Active volcano on Camiguin → **A:** *Hibok-Hibok*^D
- Q:** Active PH volcanoes per PHIVOLCS → **A:** 24 (*PHIVOLCS active list*)^{[VERIFY] D}

Common confusions

Magma vs. **lava**: magma is molten rock underground; lava is molten rock at the surface.

Crater vs. **caldera**: a crater is a small vent opening; a caldera is a large collapse basin.

Oceanic vs. **continental crust**: oceanic is thin, dense, and basaltic; continental is thick, light, and granitic.

2.3 Term 3: Typhoons, Tides, Acceleration, Energy, Light

Typhoons

KEY FACTS

- A tropical cyclone is a low-pressure system that forms over warm ocean water (about 26.5 °C or warmer). It rotates counterclockwise in the Northern Hemisphere (Coriolis effect).
- PAGASA (under DOST) categories, by sustained winds: TD ≤ 61 km/h; TS 62–88; STS 89–117; TY 118–184; Super Typhoon ≥ 185 km/h (since 2022).
- Warning signals: TCWS No. 1 to No. 5.
- Parts: the **eye** is calm with the lowest pressure; the **eyewall** has the strongest winds; then the rainbands.
- A typhoon weakens over land and mountains because it loses its warm-water fuel. The Sierra Madre shields eastern Luzon.
- About 20 tropical cyclones enter the **PAR** each year. PAR corners: 5°N 115°E, 15°N 115°E, 21°N 120°E, 25°N 120°E, 25°N 135°E, 5°N 135°E.
- A **storm surge** is an abnormal rise of the sea (Yolanda, 2013, Tacloban).

QUICK CARDS

- Q:** PH agency issuing typhoon bulletins → **A:** *PAGASA*^E
- Q:** PAGASA is under which department → **A:** *DOST*^A
- Q:** Calmest part of a typhoon → **A:** *Eye*^E
- Q:** Part with the strongest winds → **A:** *Eyewall*^A
- Q:** Spin of a typhoon in the Northern Hemisphere → **A:** *Counterclockwise*^A
- Q:** Effect that deflects winds and makes storms spin → **A:** *Coriolis effect*^D
- Q:** Why typhoons weaken after landfall → **A:** *Cut off from warm water*^A
- Q:** Mountain range shielding eastern Luzon → **A:** *Sierra Madre*^A
- Q:** Abnormal sea rise pushed by typhoon winds → **A:** *Storm surge*^E
- Q:** Year of Super Typhoon Yolanda → **A:** 2013^A
- Q:** Highest TCWS number → **A:** *Signal No. 5*^A
- Q:** Typhoons entering the PAR per year (average) → **A:** *About 20*^A
- Q:** Minimum wind for a PAGASA super typhoon → **A:** 185 km/h^D
- Q:** PAGASA wind range for “typhoon” → **A:** 118–184 km/h^D
- Q:** Eastern boundary of the PAR → **A:** 135°E^D

- Q:** Approximate minimum sea temperature for formation → **A:** *About 26.5 °C*^D

Common confusions

Eye vs. **eyewall**: the eye is calm; the eyewall around it is the most violent part.

Signal number vs. **category**: the TCWS signal is the wind threat to *your area*; the category is the storm's own strength.

Tides

KEY FACTS

- Tides are the regular rise and fall of sea level, caused mainly by the Moon's gravity and partly by the Sun's (about half as strong).
- Spring tides** (largest range) happen when the Sun, Earth, and Moon line up, at new and full moon.
- Neap tides** (smallest range) happen when the Sun and Moon are at right angles, at the quarter moons.
- Most coasts get 2 high and 2 low tides a day.
- Tidal energy is renewable. La Rance (France) is a classic tidal power plant.

QUICK CARDS

- Q:** Main cause of tides → **A:** *Moon's gravity*^E
- Q:** Highest highs and lowest lows → **A:** *Spring tides*^E
- Q:** Tide at the quarter moons → **A:** *Neap tide*^A
- Q:** Moon phases with spring tides → **A:** *New and full moon*^A
- Q:** High tides per day on most coasts → **A:** *Two*^A
- Q:** Sun's tidal effect compared with the Moon's → **A:** *About half*^D
- Q:** Famous tidal power station in France → **A:** *La Rance*^D

Common confusions

Spring vs. **neap tide**: spring = aligned, big range; neap = right angles, small range. “Spring” has nothing to do with the season.

Acceleration and Motion Graphs

KEY FACTS

- $a = \frac{v_f - v_i}{t}$, in m/s². Forces cause acceleration.
- An object accelerates when its speed *or* direction changes. Uniform circular motion is acceleration toward the centre (centripetal).
- Distance–time graph: a curve (parabola) means constant acceleration.
- Velocity–time graph: the slope is the acceleration; the area under it is the displacement; a flat line means constant velocity.
- Free-fall acceleration $g \approx 9.8 \text{ m/s}^2$.

QUICK CARDS

- Q:** SI unit of acceleration → **A:** m/s²^E
- Q:** 0 to 20 m/s in 5 s → **A:** 4 m/s²^A
- Q:** Slope of a velocity–time graph → **A:** *Acceleration*^A
- Q:** Area under a velocity–time graph → **A:** *Displacement*^D
- Q:** Horizontal line on a v–t graph → **A:** *Constant velocity (zero acceleration)*^A
- Q:** d–t graph shape for constant acceleration → **A:** *Curve (parabola)*^D
- Q:** Car rounding a curve at constant speed: accelerating? → **A:** *Yes, its direction changes*^A
- Q:** Direction of acceleration in uniform circular motion → **A:** *Toward the centre*^D
- Q:** Acceleration due to gravity → **A:** 9.8 m/s²^E
- Q:** Slowing down is called → **A:** *Deceleration*^E

Common confusions

Flat d–t line vs. flat v–t line: a flat d–t line means at rest; a flat v–t line means constant velocity.

Velocity vs. acceleration: at the top of a vertical throw $v = 0$ but $a = g$ is still acting.

Work, Energy, Power, and Hydroelectricity**KEY FACTS**

- Kinetic energy $KE = \frac{1}{2}mv^2$. Gravitational potential energy $PE = mgh$. Both are in joules.
- Work $W = Fd$ (force along the displacement). No displacement, or force $\perp$ motion, means no work.
- Power $P = W/t$, in watts. $1 \text{ hp} \approx 746 \text{ W}$.
- Mechanical energy $= KE + PE$. Energy is conserved: a falling object turns PE into KE.
- Hydroelectric plants turn the PE of dam water into KE, which turns a turbine and generator to make electricity. Examples: Angat (Bulacan; supplies most of Metro Manila's water), San Roque, Magat, Pantabangan.

QUICK CARDS

Q: Energy of motion $\rightarrow$ **A:** *Kinetic energy*^E

Q: Stored energy due to height $\rightarrow$ **A:** *Gravitational potential energy*^E

Q: Unit of work and energy $\rightarrow$ **A:** *Joule*^E

Q: Formula for work $\rightarrow$ **A:** $W = Fd$ ^E

Q: 10 N pushes a box 3 m: work $\rightarrow$ **A:** 30 J^A

Q: Pushing a wall that doesn't move: work done $\rightarrow$ **A:** Zero^A

Q: Rate of doing work $\rightarrow$ **A:** *Power*^E

Q: Unit of power $\rightarrow$ **A:** *Watt*^E

Q: $KE + PE \rightarrow$ **A:** *Mechanical energy*^A

Q: Energy that is maximum at the top of a swing $\rightarrow$ **A:** *Potential energy*^A

Q: Energy change in a falling coconut $\rightarrow$ **A:** $PE \rightarrow KE$ ^E

Q: Energy is neither created nor destroyed $\rightarrow$ **A:** *Law of Conservation of Energy*^E

Q: Doubling speed multiplies KE by $\rightarrow$ **A:** 4^D

Q: 1 horsepower is about $\rightarrow$ **A:** 746 W^D

Q: Carrying a bag while walking level at constant speed: work done on the bag $\rightarrow$ **A:** Zero (*force $\perp$ motion*)^D

Q: Bulacan dam supplying most of Metro Manila's water $\rightarrow$ **A:** *Angat Dam*^A

Q: Hydropower changes water's PE into $\rightarrow$ **A:** *Electrical energy (via KE)*^A

Common confusions

Energy vs. power: energy is the total amount (J); power is how fast it is used or transferred (W).

Work (physics) vs. effort: holding a heavy box still is tiring, but the work done on it is zero.

Properties of Light: Reflection and Refraction**KEY FACTS**

- Law of reflection: angle of incidence = angle of reflection, both measured from the normal.
- Plane mirror image: virtual, upright, same size, laterally inverted.
- **Concave** (converging) mirrors are used in flashlights, solar cookers, and shaving mirrors. **Convex** (diverging) mirrors give a wide view and are used as vehicle side mirrors and store mirrors.
- **Refraction** is light bending as its speed changes between media. Entering a denser medium (air $\rightarrow$ water), it bends toward the normal.
- A prism disperses white light into ROYGBIV; violet bends the most.
- A convex lens is a magnifier and corrects farsightedness. A concave lens corrects nearsightedness.
- Total internal reflection makes optical fibres work.

QUICK CARDS

Q: Law of reflection $\rightarrow$ **A:** *Angle of incidence = angle of reflection*^E

Q: Plane mirror image: real or virtual? $\rightarrow$ **A:** *Virtual*^E

Q: "AMBULANCE" is written in reverse because mirrors cause $\rightarrow$ **A:** *Lateral inversion*^A

Q: Mirror used for vehicle side mirrors $\rightarrow$ **A:** *Convex*^E

Q: Mirror in flashlights and solar cookers $\rightarrow$ **A:** *Concave*^A

Q: Bending of light between media $\rightarrow$ **A:** *Refraction*^E

Q: Splitting white light into colours $\rightarrow$ **A:** *Dispersion*^A

Q: Colour bent most by a prism $\rightarrow$ **A:** *Violet*^D

Q: Lens of a magnifying glass $\rightarrow$ **A:** *Convex*^E

Q: Lens that corrects nearsightedness $\rightarrow$ **A:** *Concave*^D

Q: Light going from air into water bends $\rightarrow$ **A:** *Toward the normal*^D

Q: Principle behind optical fibres $\rightarrow$ **A:** *Total internal reflection*^D

Q: Speed of light in a vacuum $\rightarrow$ **A:** $3 \times 10^8 \text{ m/s}$ ^A

Common confusions

Reflection vs. refraction: reflection bounces light back; refraction bends it as it passes through.

Concave vs. convex mirror: concave "caves in" and converges light; convex bulges out, diverges, and widens the view.

Real vs. virtual image: a real image can appear on a screen (light actually meets); a virtual image cannot.

Mnemonic: ROY G BIV: Red, Orange, Yellow, Green, Blue, Indigo, Violet (longest to shortest wavelength).

Cards in this part: 191 (E 76, A 77, D 38)

3 Grade 9

3.1 Term 1: Newton's Laws, Electricity, EM Waves, Plates

Newton's Laws of Motion

KEY FACTS

- **1st law (inertia):** an object stays at rest or in uniform motion unless an unbalanced (net) force acts on it. More mass means more inertia.
- **2nd law:** $F = ma$. Acceleration increases with net force and decreases with mass.
- **3rd law:** for every action there is an equal and opposite reaction. The two forces act on *different* objects, so they do not cancel.
- $1\text{ N} = 1\text{ kg m/s}^2$. The laws were published in the *Principia* (1687).

QUICK CARDS

- Q: Newton's law of inertia → A: *First law*^E
- Q: Tendency to resist a change in motion → A: *Inertia*^E
- Q: Quantity that measures inertia → A: *Mass*^A
- Q: Passengers lurch forward when a jeepney brakes → A: *First law (inertia)*^E
- Q: Coin drops into the glass when the card is flicked → A: *Inertia*^A
- Q: $F = ma$ is which law? → A: *Second law*^E
- Q: Net force on a 2 kg cart accelerating at 3 m/s^2 → A: 6 N ^A
- Q: Same force, double the mass: acceleration → A: *Halved*^D
- Q: Swimmer pushes water back and moves forward → A: *Third law*^E
- Q: Stepping off a bangka pushes the boat → A: *Backward (away from you)*^A
- Q: Why action and reaction don't cancel → A: *They act on different objects*^D
- Q: Book on a table: reaction to Earth pulling the book down → A: *Book pulls Earth up*^D
- Q: 1 newton in base units → A: kg m/s^2 ^D
- Q: Newton's 1687 book → A: *Principia*^D

Common confusions

Action-reaction pair vs. balanced forces: an action-reaction pair acts on two objects; balanced forces act on one object.

Mass vs. inertia: mass is the *measure* of inertia; inertia is the *property*.

Electricity and Circuits

KEY FACTS

- Current is a flow of electrons, measured in amperes (A) with an **ammeter in series**.
- Voltage (potential difference) is measured in volts (V) with a **voltmeter in parallel**. Resistance is in ohms (Ω).
- **Ohm's law** (Georg Ohm): $V = IR$.
- **Series:** one path, same current everywhere, voltages add, $R_T = R_1 + R_2$. One bulb out turns all off, and more bulbs make them dimmer.
- **Parallel:** many paths, same voltage across each branch, currents add. One bulb out leaves the others on. Homes are wired in parallel.
- Electrons flow from $-$ to $+$; conventional current flows from $+$ to $-$.
- Safety: fuses and circuit breakers, no overloaded outlets, dry hands. PH household supply is 220 V.

QUICK CARDS

- Q: Unit of current → A: *Ampere*^E

- Q: Unit of resistance → A: *Ohm*^E
- Q: Unit of potential difference → A: *Volt*^E
- Q: Meter connected in series → A: *Ammeter*^A
- Q: Meter connected in parallel → A: *Voltmeter*^A
- Q: $V = IR$ → A: *Ohm's law*^E
- Q: 12 V across 4Ω : current → A: 3 A ^A
- Q: 2Ω and 3Ω in series → A: 5Ω ^A
- Q: One bulb blows and all go off → A: *Series circuit*^E
- Q: Type of household wiring → A: *Parallel*^E
- Q: Adding bulbs in series makes them → A: *Dimmer*^A
- Q: Same across every parallel branch → A: *Voltage*^D
- Q: Same everywhere in a series circuit → A: *Current*^D
- Q: Device that cuts the circuit on overload → A: *Fuse / circuit breaker*^A
- Q: Standard household voltage in PH → A: 220 V ^A
- Q: Direction of electron flow → A: *Negative to positive terminal*^D

Common confusions

Current vs. voltage: current is the flow (A); voltage is the push (V).

Series vs. parallel: series has one path and shares voltage; parallel has many paths and each branch gets full voltage.

Fuse vs. circuit breaker: a fuse melts and must be replaced; a breaker trips and can be reset.

Mnemonic: Ohm's triangle: **V** on top, **I** and **R** below. Cover the one you need.

Electromagnetic Waves

KEY FACTS

- EMR is energy from vibrating charged particles. It travels as **transverse** waves, needs no medium, and moves at the speed of light in a vacuum.
- Order of *increasing* frequency and energy (decreasing wavelength): radio, microwave, infrared, visible, ultraviolet, X-ray, gamma.
- $v = f\lambda$. Frequency is in hertz (Hz).
- Uses: radio/TV and telecoms (radio), ovens, Wi-Fi and satellites (microwave), remotes and thermal imaging (IR), sterilising and spotting fake bills (UV), bone imaging and airport scanners (X-ray), cancer therapy and sterilising equipment (gamma).
- Hazards: UV causes sunburn, skin cancer, and cataracts. X-rays and gamma are ionizing and damage cells and DNA.
- Discoverers: **Maxwell** predicted EM waves, **Hertz** produced radio waves, **Herschel** found IR, **Röntgen** found X-rays (1895).

QUICK CARDS

- Q: EM waves are transverse or longitudinal? → A: *Transverse*^E
- Q: Lowest-frequency EM wave → A: *Radio*^E
- Q: Highest-energy EM wave → A: *Gamma*^E
- Q: EM wave used by TV remotes → A: *Infrared*^E
- Q: Causes sunburn and is used for sterilizing → A: *Ultraviolet*^E
- Q: EM wave for imaging bones → A: *X-ray*^E
- Q: Used in cancer radiotherapy → A: *Gamma rays*^A
- Q: Used by microwave ovens and Wi-Fi → A: *Microwaves*^A
- Q: Distance from crest to crest → A: *Wavelength*^E
- Q: Unit of frequency → A: *Hertz*^E
- Q: As frequency rises, wavelength → A: *Decreases*^A
- Q: Visible colour with the longest wavelength → A: *Red*^A
- Q: $f = 100\text{ Hz}$, $\lambda = 2\text{ m}$: wave speed → A: 200 m/s ^A
- Q: Discovered X-rays → A: *Wilhelm Röntgen*^A

- Q:** Predicted EM waves mathematically → **A:** *James Clerk Maxwell*^D
- Q:** First produced and detected radio waves → **A:** *Heinrich Hertz*^D
- Q:** Discovered infrared → **A:** *William Herschel*^D
- Q:** Do X-rays travel faster than radio waves in a vacuum? → **A:** *No, both travel at c*^D
- Q:** Ionizing parts of the EM spectrum → **A:** *High-energy UV, X-ray, gamma*^D

Common confusions

Frequency vs. wavelength: they are inversely related: high frequency means short wavelength.

Amplitude vs. wavelength: amplitude is the wave's height (intensity); wavelength is the crest-to-crest length.

Ionizing vs. non-ionizing: ionizing radiation (X-ray, gamma) can knock electrons off atoms and damage DNA; radio, microwave, and IR cannot.

Mnemonic: Raging Martians Invaded Venus Using X-ray Guns (radio → gamma).

Continental Drift and Plate Boundaries

KEY FACTS

- Alfred Wegener** proposed continental drift: one supercontinent, **Pangaea**, surrounded by **Panthalassa**.
- Evidence: jigsaw fit of coastlines (South America and Africa), matching rocks and mountains, fossils (*Mesosaurus*, *Glossopteris*), and old glacier deposits.
- Wegener's idea was rejected at first because he had no mechanism. **Harry Hess** later proposed seafloor spreading.
- Divergent** boundaries pull apart (mid-ocean ridges, rift valleys: Mid-Atlantic Ridge, East African Rift).
- Convergent** boundaries collide: ocean–ocean gives an island arc and trench; ocean–continent gives a trench and volcanic mountains (Andes); continent–continent gives folded mountains (Himalayas).
- Transform** boundaries slide past each other (San Andreas Fault).
- The PH sits between the **Philippine Sea Plate** (subducting at the Philippine Trench, east) and the Eurasian/Sunda Plate (subducting at the **Manila Trench**, west).

QUICK CARDS

- Q:** Proposed continental drift → **A:** *Alfred Wegener*^E
- Q:** Ancient supercontinent → **A:** *Pangaea*^E
- Q:** Ocean surrounding Pangaea → **A:** *Panthalassa*^D
- Q:** Freshwater reptile fossil in both South America and Africa → **A:** *Mesosaurus*^A
- Q:** Seed-fern fossil used as drift evidence → **A:** *Glossopteris*^A
- Q:** Why drift was rejected at first → **A:** *No mechanism explained*^D
- Q:** Proposed seafloor spreading → **A:** *Harry Hess*^D
- Q:** Boundary where plates move apart → **A:** *Divergent*^E
- Q:** Boundary where plates slide past each other → **A:** *Transform*^E
- Q:** Example of a transform boundary → **A:** *San Andreas Fault*^A
- Q:** Example of a divergent boundary → **A:** *Mid-Atlantic Ridge*^A
- Q:** Ocean–continent convergence forms → **A:** *Trench + volcanic mountains*^A
- Q:** Continent–continent convergence forms → **A:** *Folded mountains (Himalayas)*^A
- Q:** Ocean–ocean convergence forms → **A:** *Volcanic island arc*^D
- Q:** Plate that subducts at the Philippine Trench → **A:** *Philippine Sea Plate*^D
- Q:** Trench west of Luzon → **A:** *Manila Trench*^D

Common confusions

Continental drift vs. plate tectonics: drift says continents move; plate tectonics explains *how* (lithospheric plates on the asthenosphere).

Trench vs. ridge: a trench forms where plates converge (subduction); a ridge forms where they diverge.

3.2 Term 2: Geologic Time, Earth's Interior, Solar System, Biodiversity

Fossils, Dating, and the Geologic Time Scale

KEY FACTS

- Relative dating** gives the order of events. By the **law of superposition** (Nicolas Steno), the oldest undisturbed layer is at the bottom. **Index fossils** are widespread but lived for a short time.
- Absolute (radiometric) dating** gives an age in years from radioactive decay and half-life. Carbon-14 (half-life about 5 730 years) dates once-living material; uranium and potassium methods date rocks.
- Time units: eon > era > period > epoch. Earth is about 4.6 billion years old.
- Eras: Paleozoic (“ancient life”), Mesozoic (age of reptiles/dinosaurs), Cenozoic (age of mammals; today). We live in the Holocene epoch.
- The largest mass extinction was at the end of the Permian. The dinosaurs died at the end of the Cretaceous.

QUICK CARDS

- Q:** Oldest undisturbed layer is at the bottom → **A:** *Law of superposition*^E
- Q:** Formulated the law of superposition → **A:** *Nicolas Steno*^D
- Q:** Fossils best for matching rock ages → **A:** *Index fossils*^A
- Q:** Dating that gives an age in years → **A:** *Absolute (radiometric)*^E
- Q:** Time for half of a radioactive sample to decay → **A:** *Half-life*^E
- Q:** Half-life of carbon-14 → **A:** *About 5 730 years*^D
- Q:** Approximate age of Earth → **A:** *4.6 billion years*^E
- Q:** Largest division of geologic time → **A:** *Eon*^A
- Q:** “Age of reptiles” → **A:** *Mesozoic Era*^E
- Q:** Era we live in → **A:** *Cenozoic*^A
- Q:** Current epoch → **A:** *Holocene*^D
- Q:** Largest mass extinction → **A:** *End-Permian*^D

Common confusions

Relative vs. absolute dating: relative gives older or younger; absolute gives a number of years.

Eon vs. era: an eon is larger; eras are subdivisions of an eon.

Earth's Interior and Seismic Waves

KEY FACTS

- Layers: crust (thinnest), mantle (thickest), liquid outer core, solid inner core (iron–nickel). The lithosphere is rigid; the asthenosphere below it is plastic, and the plates move on it.
- P-waves** are primary, compressional, and fastest, and pass through solids and liquids. **S-waves** are secondary and transverse and pass through solids only. **Surface waves** are the most damaging.
- The S-wave shadow zone shows that the outer core is liquid.
- Boundaries: **Moho** (crust–mantle, Mohorovičić), **Gutenberg** (mantle–core). **Inge Lehmann** discovered the inner core.
- Earth's internal heat comes mainly from radioactive decay. The moving outer core generates the magnetic field.

QUICK CARDS

- Q: Fastest seismic wave → A: *P-wave*^E
 Q: Seismic wave that cannot pass through liquid → A: *S-wave*^A
 Q: Evidence that the outer core is liquid → A: *S-wave shadow zone*^D
 Q: Crust–mantle boundary → A: *Moho*^A
 Q: Discovered the inner core → A: *Inge Lehmann*^D
 Q: Thickest layer → A: *Mantle*^E
 Q: Thinnest layer → A: *Crust*^E
 Q: Plastic layer the plates ride on → A: *Asthenosphere*^A
 Q: Main metals of the core → A: *Iron and nickel*^E
 Q: Layer that generates Earth's magnetic field → A: *Outer core*^A
 Q: Main source of Earth's internal heat → A: *Radioactive decay*^A

Common confusions

P-wave vs. **S-wave**: P waves are push–pull and travel through anything; S waves shake side to side through solids only.

Lithosphere vs. **asthenosphere**: the lithosphere is rigid (crust + top mantle); the asthenosphere is soft and flowing.

Crust vs. **lithosphere**: the lithosphere includes the crust *plus* the uppermost mantle.

Mnemonic: P = Primary, Push–pull, Passes all. S = Secondary, Shake, Solids only.

The Solar System, Small Bodies, and Space Technology

KEY FACTS

- The **nebular hypothesis** says the Solar System formed about 4.6 billion years ago from a spinning cloud of gas and dust.
- In 2006 the IAU made Pluto a dwarf planet. The dwarf planets are Ceres, Pluto, Eris, Haumea, and Makemake.
- Asteroids** are rocky and mostly lie between Mars and Jupiter. **Comets** are icy, with tails pointing away from the Sun. Short-period comets come from the Kuiper belt, long-period ones from the Oort cloud.
- A meteoroid is in space, a meteor is the streak of light in the sky, and a meteorite is what lands.
- Meteor showers happen when Earth crosses comet debris: Perseids (Aug), Orionids (Oct), Leonids (Nov), Geminids (Dec, from asteroid Phaethon).
- Technology: Galileo first used a telescope for astronomy. Telescopes are refracting (lenses) or reflecting (mirrors). Hubble launched in 1990; JWST (infrared) in 2021. Probes: Voyager 1, New Horizons (Pluto, 2015).
- PH space efforts: **PhilSA**; **Diwata-1** was the first PH microsatellite (2016).

QUICK CARDS

- Q: Theory that the Solar System formed from a spinning gas cloud → A: *Nebular hypothesis*^E
 Q: Year Pluto was reclassified → A: *2006*^A
 Q: Body that reclassified Pluto → A: *IAU*^A
 Q: Dwarf planet in the asteroid belt → A: *Ceres*^A
 Q: Location of the asteroid belt → A: *Between Mars and Jupiter*^E
 Q: Icy body with a tail away from the Sun → A: *Comet*^E
 Q: Streak of light in the sky → A: *Meteor*^E
 Q: Space rock that reaches the ground → A: *Meteorite*^E
 Q: Cause of meteor showers → A: *Earth passing through comet debris*^A
 Q: August meteor shower → A: *Perseids*^A
 Q: December shower from asteroid Phaethon → A: *Geminids*^D

- Q: Orbital period of Halley's Comet → A: *About 76 years*^A
 Q: Source region of long-period comets → A: *Oort cloud*^D
 Q: Region beyond Neptune that includes Pluto → A: *Kuiper belt*^A
 Q: Largest planet → A: *Jupiter*^E
 Q: Hottest planet → A: *Venus*^A
 Q: Telescope that uses mirrors → A: *Reflecting telescope*^A
 Q: First to observe Jupiter's moons with a telescope → A: *Galileo Galilei*^E
 Q: Infrared space telescope launched in 2021 → A: *James Webb (JWST)*^A
 Q: First spacecraft to enter interstellar space → A: *Voyager 1*^D
 Q: Probe that flew past Pluto in 2015 → A: *New Horizons*^D
 Q: Philippine Space Agency → A: *PhilSA*^A
 Q: First Philippine microsatellite → A: *Diwata-1*^D

Common confusions

Meteoroid vs. **meteor**: meteoroid (in space) → meteor (light in the sky) → meteorite (on the ground).

Asteroid vs. **comet**: asteroids are rocky with no tail; comets are icy and grow tails near the Sun.

Refracting vs. **reflecting telescope**: refracting telescopes use lenses; reflecting telescopes use mirrors (all large modern ones).

Mnemonic: My Very Educated Mother Just Served Us Nachos (Mercury → Neptune).

Biodiversity and Endangered Species

KEY FACTS

- Biodiversity** has three levels: genetic, species, and ecosystem. High biodiversity gives a more stable, resilient ecosystem with more alternative food sources and pathways.
- IUCN Red List: EX, EW, **CR**, **EN**, **VU**, NT, LC (plus DD). “Threatened” means CR, EN, or VU.
- Critically endangered PH species include the Philippine eagle, the tamaraw (Mindoro), the Philippine cockatoo (*katala*), and the Philippine pangolin (IUCN: CR) ^[VERIFY].
- Main threats (HIPPO): habitat loss, invasive species (janitor fish, golden apple snail), pollution, human population, overharvesting.
- Protection: in-situ (protected areas such as Tubbataha Reefs Natural Park) and ex-situ (zoos, seed banks). Laws: NIPAS Act (RA 7586) and the Wildlife Act (RA 9147).

QUICK CARDS

- Q: Variety of life in an area → A: *Biodiversity*^E
 Q: Three levels of biodiversity → A: *Genetic, species, ecosystem*^A
 Q: Species found naturally in only one place → A: *Endemic species*^E
 Q: IUCN category one step riskier than Endangered → A: *Critically Endangered*^A
 Q: IUCN categories that count as “threatened” → A: *CR, EN, VU*^D
 Q: IUCN status of the Philippine eagle → A: *Critically Endangered*^A
 Q: Dwarf buffalo endemic to Mindoro → A: *Tamaraw*^E
 Q: Local name of the Philippine cockatoo → A: *Katala*^A
 Q: Local name of the whale shark → A: *Butanding*^E
 Q: Leading cause of biodiversity loss → A: *Habitat loss*^A
 Q: Invasive fish in Laguna de Bay and Marikina River → A: *Janitor fish*^A
 Q: Invasive snail that pests rice fields → A: *Golden apple snail (golden kuhol)*^A
 Q: Conservation within the natural habitat → A: *In-situ*^D

- Q: Conservation outside the habitat (zoos, seed banks) → A: *Ex-situ*^D
- Q: UNESCO reef park in the Sulu Sea → A: *Tubbataha Reefs*^A
- Q: Law creating the protected-areas system → A: *NIPAS Act (RA 7586)*^D
- Q: PH Wildlife Act → A: *RA 9147*^D
- Q: Why high biodiversity keeps ecosystems stable → A: *Alternative food sources and pathways*^D

Common confusions

Extinct vs. **extinct in the wild**: extinct means no individuals left anywhere; EW means they survive only in captivity.

Endemic vs. **native**: endemic means found *only* there; native means naturally there but possibly elsewhere too.

In-situ vs. **ex-situ**: in-situ is in the natural habitat; ex-situ is outside it.

Mnemonic: HIPPO: Habitat loss, Invasive species, Pollution, Population (human), Overharvesting.

Philippine Ecosystems and Human Impact

KEY FACTS

- **Tropical rainforest**: high rainfall and biodiversity, with layers (emergent, canopy, understory, floor).
- **Mangroves** are salt-tolerant intertidal trees with prop roots and pneumatophores (breathing roots). They are fish nurseries and buffer coasts against surges.
- **Estuary**: where a river meets the sea, with brackish and very productive water.
- **Swamp**: a tree-dominated wetland (e.g., Agusan Marsh).
- **Coral reefs** are built by coral polyps (animals) that host algae (zooxanthellae). Heat stress causes bleaching. The PH is in the **Coral Triangle**.
- Human threats: deforestation and *kaingin*, pollution, dynamite and cyanide fishing, invasive species, and converting mangroves to fishponds.

QUICK CARDS

- Q: Mix of fresh and salt water → A: *Brackish*^E
- Q: Where a river meets the sea → A: *Estuary*^E
- Q: Breathing roots of mangroves → A: *Pneumatophores*^D
- Q: Local term for slash-and-burn farming → A: *Kaingin*^E
- Q: Animals that build coral reefs → A: *Coral polyps*^A
- Q: Algae living inside coral tissue → A: *Zooxanthellae*^D
- Q: Corals turning white from heat stress → A: *Coral bleaching*^A
- Q: Marine biodiversity hotspot that includes the PH → A: *Coral Triangle*^A
- Q: Large wetland in Agusan del Sur → A: *Agusan Marsh*^A
- Q: Coastal protection role of mangroves → A: *Buffer against surges and erosion*^A
- Q: Dense “roof” layer of a rainforest → A: *Canopy*^E
- Q: Two destructive fishing methods → A: *Dynamite and cyanide fishing*^E
- Q: Example of an abiotic factor → A: *Sunlight, water, temperature*^E
- Q: Major historic cause of PH mangrove loss → A: *Conversion to fishponds (main historic cause of PH mangrove loss)*^{[VERIFY] D}

Common confusions

Biotic vs. **abiotic**: biotic factors are living (plants, animals); abiotic are nonliving (light, water, soil).

Swamp vs. **marsh**: swamps are dominated by trees; marshes by grasses and reeds.

Estuary vs. **mangrove**: an estuary is a *place* (river mouth); a mangrove is a *forest type* often found there.

3.3 Term 3: DNA and Mutations, Chemical Bonding

DNA, Genes, Chromosomes, and Mutations

KEY FACTS

- DNA (deoxyribonucleic acid) is a **double helix** (Watson and Crick, 1953), built using **Rosalind Franklin’s** X-ray image (Photo 51).
- A nucleotide is phosphate + deoxyribose + base. Bases pair **A–T** and **G–C**. The sugar–phosphate backbone forms the sides; the base pairs form the rungs.
- A gene is a DNA segment that codes for a protein or trait. A chromosome is coiled DNA. Humans have 23 pairs (22 autosome pairs + XX or XY).
- Replication is semi-conservative. Helicase unzips the DNA and DNA polymerase builds the new strand.
- **Mutation** is a change in DNA or chromosomes. Gene mutations: substitution, insertion, deletion (frameshift). Chromosomal mutations: deletion, duplication, inversion, translocation, and extra or missing chromosomes (trisomy 21 is Down syndrome).
- Mutagens: radiation (UV, X-ray), chemicals (tobacco smoke), and infectious agents (some viruses). Effects can be harmful, beneficial, or neutral.

QUICK CARDS

- Q: DNA stands for → A: *Deoxyribonucleic acid*^E
- Q: Shape of DNA → A: *Double helix*^E
- Q: Proposed the double-helix model (1953) → A: *Watson and Crick*^E
- Q: Scientist behind Photo 51 → A: *Rosalind Franklin*^A
- Q: Base that pairs with adenine → A: *Thymine*^E
- Q: Base that pairs with guanine → A: *Cytosine*^E
- Q: Sugar in DNA → A: *Deoxyribose*^A
- Q: Three parts of a nucleotide → A: *Phosphate, sugar, base*^A
- Q: DNA segment coding for a trait → A: *Gene*^E
- Q: Chromosome pairs in a human body cell → A: *23*^E
- Q: Sex chromosomes of a male → A: *XY*^E
- Q: Enzyme that unzips DNA → A: *Helicase*^D
- Q: Each new DNA keeps one old strand → A: *Semi-conservative replication*^D
- Q: Change in the DNA sequence → A: *Mutation*^E
- Q: Agent that causes mutation → A: *Mutagen*^A
- Q: Extra copy of chromosome 21 → A: *Down syndrome*^A
- Q: Mutation with no effect → A: *Neutral (silent)*^A
- Q: Insertion or deletion that shifts the reading frame → A: *Frameshift mutation*^D
- Q: Disorder from one base substitution in the haemoglobin gene → A: *Sickle-cell anaemia*^D
- Q: If 20% of the bases are A, the percentage of G → A: *30%*^D

Common confusions

DNA vs. **gene**: a gene is one short section of a DNA molecule; a chromosome is one long DNA molecule packed with proteins.

Gene mutation vs. **chromosomal mutation**: a gene mutation changes a few bases; a chromosomal mutation changes large segments or whole chromosomes.

DNA vs. **RNA**: DNA has deoxyribose, T, and two strands; RNA has ribose, U, and one strand.

Chemical Change and Valence Electrons

KEY FACTS

- A chemical change breaks old bonds and forms new ones, making a new substance.

- Evidence tests: CO_2 turns **limewater** milky (a CaCO_3 precipitate). AgNO_3 + salt solution gives a white AgCl precipitate.
- **Valence electrons** are outer-shell electrons and take part in bonding. Oxygen (Group 16) has 6.
- The **octet rule**: atoms tend to gain, lose, or share electrons to have 8 valence electrons. Lewis dot diagrams show valence electrons.

QUICK CARDS

- Q: Electrons in the outermost shell → A: *Valence electrons*^E
 Q: Valence electrons of oxygen → A: 6^E
 Q: Atoms tend to reach 8 valence electrons → A: *Octet rule*^E
 Q: Electron-dot diagrams are named after → A: *Gilbert N. Lewis*^A
 Q: Test for CO_2 → A: *Limewater turns milky*^A
 Q: White precipitate from silver nitrate + salt water → A: *Silver chloride*^D

Ionic, Covalent, and Metallic Bonding

KEY FACTS

- **Ionic** bonds form when a metal transfers electrons to a non-metal: $\text{Na} \longrightarrow \text{Na}^+ + \text{e}^-$ and $\text{Cl} + \text{e}^- \longrightarrow \text{Cl}^-$. The ions form a **crystal lattice** that has a high melting point, is brittle, and conducts when molten or dissolved.
- Formulas: NaCl sodium chloride, MgO magnesium oxide, KCl potassium chloride, MgCl_2 magnesium chloride.
- **Covalent** bonds form when nonmetals share electrons, making molecules with low melting points that are poor conductors: H_2O , CO_2 ($\text{O}=\text{C}=\text{O}$), NH_3 .
- **Metallic** bonds hold metal cations in a **sea of electrons**, so metals conduct and are malleable, ductile, and shiny.

QUICK CARDS

- Q: Positively charged ion → A: *Cation*^E
 Q: Negatively charged ion → A: *Anion*^E
 Q: How sodium becomes Na^+ → A: *Loses 1 electron*^A
 Q: Ion charge of magnesium → A: $2+$ ^A
 Q: Bond formed by electron transfer → A: *Ionic*^E
 Q: Bond formed by sharing electrons → A: *Covalent*^E
 Q: Bond within metals → A: *Metallic*^E
 Q: Model of metallic bonding → A: *Sea of electrons*^A
 Q: Formula of magnesium chloride → A: MgCl_2 ^A
 Q: Name of MgO → A: *Magnesium oxide*^E
 Q: Formula of ammonia → A: NH_3 ^A
 Q: Structure of ionic compounds → A: *Crystal lattice*^A
 Q: High melting point, brittle, conducts when dissolved → A: *Ionic compound*^A
 Q: Bonds in a CO_2 molecule → A: *Two double covalent bonds*^D
 Q: Why molten NaCl conducts electricity → A: *Its ions are free to move*^D
 Q: Why metals can be hammered into sheets → A: *Cations slide within the electron sea*^D
 Q: Bonding in diamond → A: *Covalent (network)*^D

Common confusions

Ionic vs. **covalent**: ionic transfers electrons (metal + nonmetal); covalent shares them (nonmetal + nonmetal).

Cation vs. **anion**: a cation *lost* electrons (+); an anion *gained* electrons (-).

Molecule vs. **crystal lattice**: covalent compounds form separate molecules; ionic compounds form a continuous lattice (no single "molecule").

Mnemonic: Cations are "pawsitive" (the *t* looks like +); Anion = A Negative ION.

Cards in this part: 186 (E 67, A 70, D 49)

4 Grade 10

Scope note: DepEd's original phase-in schedule puts Grade 10 MATATAG in SY 2027–2028, but a Grade 10 MATATAG Three-Term BOW exists for SY 2026–2027. The core below follows that BOW. Topics found only in the older K-12 MELCs are in the appendix at the end of this section. Confirm with your school which Grade 10 curriculum applies this year. [\[VERIFY\]](#)

4.1 Term 1: Chemical Reactions, Homeostasis, Evolution

Evidence of Reactions; Acids, Bases, and Indicators

KEY FACTS

- Signs of a chemical reaction: colour change, precipitate, gas (bubbles), odour, temperature change (and sometimes light).
- Acids release H^+ and taste sour. Bases release OH^- , taste bitter, and feel slippery. Acid + base $\rightarrow$ salt + water (**neutralization**).
- Common acids: HCl (stomach acid, muriatic acid), acetic acid (vinegar/*suka*), citric acid (calamansi). Common bases: NaOH (lye, caustic soda), $Mg(OH)_2$ (antacid), ammonia (cleaners). Salt solution: saline (NaCl).
- Indicators: phenolphthalein is colourless in acid and pink in base. Methyl orange is red in acid and yellow in base. Red cabbage is red in acid and green-yellow in base. Turmeric (*tuyang dilaw*) is yellow, turning reddish-brown in base.

QUICK CARDS

- Q: Five signs of a chemical reaction $\rightarrow$ **A:** *Colour, gas, precipitate, odour, temperature change*^A
- Q: Insoluble solid formed from two solutions $\rightarrow$ **A:** *Precipitate*^E
- Q: Phenolphthalein in a base $\rightarrow$ **A:** *Pink*^A
- Q: Phenolphthalein in an acid $\rightarrow$ **A:** *Colourless*^A
- Q: Acid in the stomach $\rightarrow$ **A:** *Hydrochloric acid*^E
- Q: Acid in vinegar $\rightarrow$ **A:** *Acetic acid*^E
- Q: Common name of NaOH $\rightarrow$ **A:** *Lye (caustic soda)*^A
- Q: Chemical name of baking soda $\rightarrow$ **A:** *Sodium bicarbonate ($NaHCO_3$)*^A
- Q: Acid + base gives $\rightarrow$ **A:** *Salt + water*^E
- Q: Base in milk of magnesia antacid $\rightarrow$ **A:** *Magnesium hydroxide*^A
- Q: Turmeric indicator in a base turns $\rightarrow$ **A:** *Reddish-brown*^D

Types of Chemical Reactions

KEY FACTS

- Combination (synthesis):** $A + B \rightarrow AB$, e.g. $2Mg + O_2 \rightarrow 2MgO$.
- Decomposition:** $AB \rightarrow A + B$, e.g. $2H_2O_2 \rightarrow 2H_2O + O_2$.
- Single replacement:** $A + BC \rightarrow AC + B$, e.g. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ (acid + metal gives hydrogen).
- Double replacement:** $AB + CD \rightarrow AD + CB$, e.g. $AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$.
- Combustion:** fuel + $O_2 \rightarrow CO_2 + H_2O$, e.g. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$.
- Acid + carbonate gives salt + water + CO_2 (vinegar + baking soda).
- Gas tests: H_2 gives a “squeaky pop” with a burning splint; O_2 relights a glowing splint.
- Rusting needs iron, oxygen, and water. Galvanizing (a zinc coating) prevents it.

QUICK CARDS

- Q: $A + B \rightarrow AB \rightarrow$ **A:** *Combination (synthesis)*^E
- Q: $AB \rightarrow A + B \rightarrow$ **A:** *Decomposition*^E
- Q: $A + BC \rightarrow AC + B \rightarrow$ **A:** *Single replacement*^E
- Q: $AB + CD \rightarrow AD + CB \rightarrow$ **A:** *Double replacement*^E
- Q: Burning LPG is what type of reaction? $\rightarrow$ **A:** *Combustion*^E

- Q: Gas from acid + metal $\rightarrow$ **A:** *Hydrogen*^A
- Q: Gas from vinegar + baking soda $\rightarrow$ **A:** *Carbon dioxide*^E
- Q: Products of complete hydrocarbon combustion $\rightarrow$ **A:** *CO_2 + water*^A
- Q: Test for hydrogen $\rightarrow$ **A:** *Squeaky pop with a burning splint*^D
- Q: Test for oxygen $\rightarrow$ **A:** *Relights a glowing splint*^D
- Q: Three things needed for rusting $\rightarrow$ **A:** *Iron, oxygen, water*^A
- Q: Zinc coating that protects iron $\rightarrow$ **A:** *Galvanizing*^D

Common confusions

Single vs. double replacement: single: an element swaps into a compound. Double: two compounds swap partners (often making a precipitate).

Physical vs. chemical change: physical changes form or state only (melting, dissolving); chemical changes make a new substance.

Chemical Equations and Conservation of Mass

KEY FACTS

- Antoine Lavoisier:** mass is neither created nor destroyed in a reaction. Atoms only rearrange.
- Write a word equation, then a formula equation, then balance it by changing **coefficients only**, never subscripts.
- $2H_2 + O_2 \rightarrow 2H_2O$; $N_2 + 3H_2 \rightarrow 2NH_3$.

QUICK CARDS

- Q: Law of conservation of mass $\rightarrow$ **A:** *Antoine Lavoisier*^E
- Q: What you may change when balancing $\rightarrow$ **A:** *Coefficients only*^A
- Q: Balance $H_2 + O_2 \rightarrow H_2O \rightarrow$ **A:** *2, 1, 2*^A
- Q: Balance $N_2 + H_2 \rightarrow NH_3 \rightarrow$ **A:** *1, 3, 2*^D
- Q: Sealed flask reaction: mass after vs. before $\rightarrow$ **A:** *The same*^E

Common confusions

Coefficient vs. subscript: a coefficient ($2H_2O$) counts molecules; a subscript (H_2O) is part of the identity, so changing it changes the substance.

Rates of Reaction; Exothermic and Endothermic

KEY FACTS

- Collision theory:** particles must collide with enough energy (the activation energy) and the right orientation.
- Rate factors: temperature, concentration, surface area (particle size), and catalysts.
- A **catalyst** speeds a reaction by lowering the activation energy and is not used up. Enzymes are biological catalysts.
- Applications: refrigeration slows spoilage, the fire triangle (heat, fuel, oxygen), and painting or galvanizing slows corrosion.
- Exothermic** reactions release heat (combustion, respiration, neutralization). **Endothermic** reactions absorb heat (photosynthesis, cold packs).

QUICK CARDS

- Q: Four factors affecting reaction rate $\rightarrow$ **A:** *Temperature, concentration, surface area, catalyst*^A
- Q: Why we refrigerate food $\rightarrow$ **A:** *Low temperature slows reactions*^E
- Q: Speeds a reaction without being used up $\rightarrow$ **A:** *Catalyst*^E
- Q: Biological catalysts $\rightarrow$ **A:** *Enzymes*^E
- Q: A catalyst works by lowering the $\rightarrow$ **A:** *Activation energy*^D

- Q: Why powdered reactants react faster than lumps → A: *Greater surface area*^A
- Q: Reaction that releases heat → A: *Exothermic*^E
- Q: Cold pack or photosynthesis → A: *Endothermic*^A
- Q: Three sides of the fire triangle → A: *Heat, fuel, oxygen*^E
- Q: Particles must collide with enough energy and correct orientation → A: *Collision theory*^D

Common confusions

Exothermic vs. endothermic: exo means heat *exits*, so the surroundings get warmer; endo means heat *enters*, so the surroundings get colder.

Homeostasis and Feedback

KEY FACTS

- **Homeostasis** (term by Walter Cannon) is a stable internal balance. Indicators include body temperature (about 37°C), blood glucose, and blood pressure (about 120/80 mmHg).
- A feedback loop runs receptor → control centre (brain/hypothalamus) → effector.
- **Negative feedback** reverses a change and is the most common type: thermoregulation, and blood sugar control by **insulin** (lowers) and **glucagon** (raises), both from the pancreas.
- **Positive feedback** amplifies a change: childbirth contractions (oxytocin), blood clotting.
- When hot: sweating and vasodilation. When cold: shivering and vasoconstriction.

QUICK CARDS

- Q: Maintaining a stable internal environment → A: *Homeostasis*^E
- Q: Coined “homeostasis” → A: *Walter Cannon*^D
- Q: Normal body temperature → A: *About 37°C*^E
- Q: Brain part that acts as the body’s thermostat → A: *Hypothalamus*^A
- Q: Hormone that lowers blood sugar → A: *Insulin*^E
- Q: Hormone that raises blood sugar → A: *Glucagon*^A
- Q: Organ that secretes insulin → A: *Pancreas*^E
- Q: Feedback that reverses a change → A: *Negative feedback*^E
- Q: Example of positive feedback → A: *Childbirth contractions*^A
- Q: Blood clotting uses which feedback? → A: *Positive*^D
- Q: Body’s responses to overheating → A: *Sweating, vasodilation*^A
- Q: Shivering is a response to → A: *Cold*^E
- Q: Part of the loop that detects change → A: *Receptor*^A
- Q: Part of the loop that carries out the response → A: *Effector*^D
- Q: Typical normal adult blood pressure → A: *120/80 mmHg*^A
- Q: Disease of poor blood-sugar control → A: *Diabetes mellitus*^E

Common confusions

Negative vs. positive feedback: negative brings things back to normal; positive pushes further from normal until an endpoint.

Insulin vs. glucagon: insulin puts sugar into cells (lowers it); glucagon releases stored glucose (raises it).

Vasodilation vs. vasoconstriction: dilation widens vessels to lose heat; constriction narrows them to keep heat.

Mechanisms and Evidence of Evolution

KEY FACTS

- **Charles Darwin:** natural selection; *On the Origin of Species* (1859); HMS *Beagle*; Galápagos finches. **Alfred Russel Wallace** reached the same idea independently.

- **Lamarck:** use and disuse, inheritance of acquired traits (now rejected).
- Natural selection: overproduction, variation, competition, survival and reproduction of the fittest, then inheritance.
- Key concepts: variation, heredity, isolation (leading to new species), selection, adaptation.
- Evidence: fossils (transitional *Archaeopteryx*), biogeography, and comparative morphology: homologous structures (same origin), analogous structures (same function), vestigial structures (appendix, tailbone).
- Classic examples: the peppered moth and antibiotic resistance.

QUICK CARDS

- Q: Primary mechanism of evolution → A: *Natural selection*^E
- Q: Darwin’s 1859 book → A: *On the Origin of Species*^E
- Q: Ship Darwin sailed on → A: *HMS Beagle*^A
- Q: Islands of Darwin’s finches → A: *Galápagos*^E
- Q: Co-discoverer of natural selection → A: *Alfred Russel Wallace*^A
- Q: Theory of use and disuse → A: *Lamarck*^E
- Q: Coined “survival of the fittest” → A: *Herbert Spencer*^D
- Q: Same structure, different function (human arm, bat wing) → A: *Homologous*^E
- Q: Same function, different origin (bird vs. insect wing) → A: *Analogous*^A
- Q: Reduced, nearly functionless organ (appendix) → A: *Vestigial*^E
- Q: Reptile–bird transitional fossil → A: *Archaeopteryx*^A
- Q: Study of where species are distributed → A: *Biogeography*^A
- Q: Bacteria surviving antibiotics is an example of → A: *Natural selection*^A
- Q: Moth that darkened in industrial England → A: *Peppered moth*^D
- Q: Separation of populations leading to new species → A: *Isolation* → *speciation*^D
- Q: Inherited trait that improves survival → A: *Adaptation*^E
- Q: “Fittest” means → A: *Most surviving offspring*^D

Common confusions

Darwin vs. Lamarck: Darwin: individuals with existing variations survive. Lamarck: traits gained during life are passed on (wrong).

Homologous vs. analogous: homologous means same ancestry; analogous means same job but different ancestry.

Evolution vs. natural selection: evolution is the change over generations; natural selection is the main *mechanism*.

4.2 Term 2: Populations, Biotechnology, Climate

Carrying Capacity and Population Growth

KEY FACTS

- **Carrying capacity** (*K*) is the largest population an environment can sustain.
- Limiting factors include food, water, shelter, mates, and space.
- Density-dependent factors include competition, disease, and predation. Density-independent factors include typhoons, fires, and drought.
- Exponential growth gives a J-curve. Logistic growth gives an S-curve that levels off at *K*, so growth slows as the population nears *K*.
- Population changes through births, deaths, immigration (in), and emigration (out).

QUICK CARDS

- Q: Largest population an environment can support → **A:** *Carrying capacity*^E
- Q: Symbol for carrying capacity → **A:** K ^A
- Q: Growth with unlimited resources → **A:** *J-curve (exponential)*^A
- Q: Growth that levels off at K → **A:** *S-curve (logistic)*^A
- Q: A factor such as food that restricts growth → **A:** *Limiting factor*^E
- Q: Near K , population growth → **A:** *Slows down*^E
- Q: Moving into a population → **A:** *Immigration*^E
- Q: A typhoon kills animals regardless of crowding → **A:** *Density-independent*^D
- Q: Disease spreads faster in crowded populations → **A:** *Density-dependent*^D
- Q: Purpose of a closed fishing season → **A:** *Let fish stocks recover*^A

Common confusions

Immigration vs. **emigration**: immigration means in; emigration means exit.

Density-dependent vs. **independent**: dependent factors get stronger with crowding; independent factors hit regardless of crowding.

Biotechnology

KEY FACTS

- **Biotechnology** uses living organisms or their processes to make products.
- Traditional biotechnology is fermentation: bread (yeast), vinegar/suka, soy sauce/toyo, patis, bagoong, kesong puti, basi, and nata de coco (bacterial cellulose).
- Modern biotechnology uses genetic engineering (GMOs), for example Golden Rice (beta-carotene), Bt corn (an insect-resistance gene from *Bacillus thuringiensis*), and human insulin made by bacteria.
- Reproductive technology and cloning: IVF (Louise Brown, 1978, first “test-tube baby”); Dolly the sheep (1996, first mammal cloned from an adult cell). CRISPR is a gene-editing tool.
- PH research: IRRI and PhilRice. The ethical debate covers safety, ecology, access, and religion.

QUICK CARDS

- Q: Microbe in bread and alcohol fermentation → **A:** *Yeast*^E
- Q: Fermented coconut-water gel → **A:** *Nata de coco*^E
- Q: Bacterium that makes nata de coco → **A:** *Acetobacter (Komagataeibacter) xylinus*^D
- Q: Fresh white cheese from carabao’s milk → **A:** *Kesong puti*^A
- Q: Ilocano sugarcane wine → **A:** *Basi*^A
- Q: Patis is traditional or modern biotechnology? → **A:** *Traditional (fermentation)*^E
- Q: Rice engineered to make beta-carotene → **A:** *Golden Rice*^E
- Q: “Bt” in Bt corn stands for → **A:** *Bacillus thuringiensis*^D
- Q: Bt corn resists → **A:** *Insect pests (corn borer)*^A
- Q: Organism with deliberately altered DNA → **A:** *GMO*^E
- Q: Human insulin is mass-produced using → **A:** *Genetically engineered bacteria*^A
- Q: First mammal cloned from an adult cell → **A:** *Dolly the sheep*^A
- Q: First IVF baby (1978) → **A:** *Louise Brown*^D
- Q: Gene-editing tool (Nobel Prize 2020) → **A:** *CRISPR-Cas9*^D
- Q: International rice institute in Los Baños → **A:** *IRRI*^A

Common confusions

Traditional vs. **modern biotech**: traditional uses whole organisms naturally (fermentation); modern changes DNA directly.

GMO vs. **hybrid**: a hybrid comes from crossbreeding (selective breeding); a GMO has genes inserted in a lab.

Cloning vs. **IVF**: cloning copies one parent’s DNA; IVF is fertilization outside the body, with two parents.

Global Climate, ENSO, and Sustainability

KEY FACTS

- The **greenhouse effect** is natural. Extra greenhouse gases (CO₂, CH₄, N₂O, water vapour, fluorinated gases) enhance it and cause global warming.
- Evidence: rising temperatures, sea-level rise (thermal expansion and melting land ice), shrinking glaciers, ocean acidification, and stronger extremes. The Keeling Curve tracks rising CO₂ at Mauna Loa.
- Global warming so far is about 1.1 °C above pre-industrial levels (IPCC AR6, 2011–2020) ^[VERIFY].
- Agreements: Kyoto Protocol (1997) and the Paris Agreement (2015; well below 2 °C, aiming for 1.5 °C). The Montreal Protocol bans CFCs (ozone).
- **ENSO**: El Niño is a warm central/eastern Pacific, bringing below-normal rain and drought to the PH. La Niña is the cool phase, bringing above-normal rain.
- PH laws: Climate Change Act (RA 9729) and Renewable Energy Act (RA 9513). Renewables: geothermal, hydro, solar, wind (Bangui, Ilocos Norte), biomass.
- **Mitigation** reduces emissions; **adaptation** adjusts to the impacts.

QUICK CARDS

- Q: Main human-made greenhouse gas → **A:** *Carbon dioxide*^E
- Q: Greenhouse gas from rice paddies, livestock, and landfills → **A:** *Methane*^A
- Q: Natural warming by atmospheric gases → **A:** *Greenhouse effect*^E
- Q: Graph of rising CO₂ at Mauna Loa → **A:** *Keeling Curve*^D
- Q: 2015 pact to limit warming to well below 2 °C → **A:** *Paris Agreement*^A
- Q: UN scientific body assessing climate change → **A:** *IPCC*^A
- Q: Treaty that phased out CFCs → **A:** *Montreal Protocol*^D
- Q: Warming of the central–eastern equatorial Pacific → **A:** *El Niño*^E
- Q: El Niño’s effect on PH rainfall → **A:** *Below normal (drought)*^A
- Q: La Niña’s effect on PH rainfall → **A:** *Above normal*^A
- Q: ENSO stands for → **A:** *El Niño–Southern Oscillation*^A
- Q: Seawater becoming more acidic from absorbed CO₂ → **A:** *Ocean acidification*^A
- Q: Two causes of sea-level rise → **A:** *Thermal expansion + melting land ice*^D
- Q: PH Climate Change Act → **A:** *RA 9729*^D
- Q: PH Renewable Energy Act → **A:** *RA 9513*^D
- Q: Wind farm in Ilocos Norte → **A:** *Bangui Wind Farm*^A
- Q: Approximate global warming so far → **A:** *About 1.1 °C of global warming* ^{[VERIFY] D}

Common confusions

Greenhouse effect vs. **ozone hole**: the greenhouse effect traps heat (CO₂); the ozone hole lets in more UV (CFCs). They are different problems.

El Niño vs. **La Niña**: in the PH, El Niño means dry and La Niña means wet.

Mitigation vs. **adaptation**: mitigation cuts the cause (emissions); adaptation copes with the effects (sea walls, drought-tolerant crops).

Global warming vs. **climate change**: warming is the temperature rise; climate change is the wider set of changes it drives.

4.3 Term 3: Plate Tectonics, Projectiles, Momentum, Electricity Supply

Plate Tectonics: Drivers, Measurement, Mountain Building

KEY FACTS

- Plate motion is measured with **GPS** (mm/yr precision), satellite methods, and seismic monitoring. Plates move a few cm per year, about as fast as fingernails grow.
- Driving forces: mantle convection, **ridge push**, and **slab pull** (the strongest). The plates ride on the plastic asthenosphere.
- Subduction of an oceanic plate makes a trench, a volcanic arc, and deep quakes along the Wadati–Benioff zone.
- The **Himalayas** come from the India–Eurasia (continent–continent) collision. The **Andes** come from the Nazca Plate subducting under South America.
- PH setting: the **Philippine Mobile Belt**, squeezed between the Philippine Sea Plate and the Eurasian (Sunda) Plate. Mt. Apo is the highest PH peak; the Sierra Madre is the longest PH range.
- The Philippine Sea Plate moves at about 7 cm to 8 cm per year (Philippine Sea Plate speed) ^[VERIFY]. At 8 cm/yr for 50 million years, that is about 4000 km.

QUICK CARDS

- Q: Satellite system that measures plate motion → A: *GPS*^E
 Q: Strongest force driving plate motion → A: *Slab pull*^D
 Q: Force from elevated mid-ocean ridges → A: *Ridge push*^A
 Q: Heat-driven circulation in the mantle → A: *Mantle convection*^E
 Q: Sloping quake zone along a subducting slab → A: *Wadati–Benioff zone*^D
 Q: Plates whose collision built the Himalayas → A: *Indian and Eurasian*^A
 Q: Plate subducting beneath South America → A: *Nazca Plate*^A
 Q: Highest peak on Earth → A: *Mt. Everest*^E
 Q: Highest mountain in the PH → A: *Mt. Apo*^E
 Q: Longest mountain range in the PH → A: *Sierra Madre*^A
 Q: Subducted plate melting forms → A: *A volcanic arc*^A
 Q: Plate speeds are similar to → A: *Fingernail growth*^A
 Q: Mobile zone the PH archipelago sits on → A: *Philippine Mobile Belt*^D
 Q: 8 cm/yr for 50 million years → A: *About 4000 km*^D

Common confusions

Ridge push vs. **slab pull**: ridge push shoves from the spreading ridge; slab pull drags from the sinking edge (stronger).

Andes vs. **Himalayas**: the Andes are ocean–continent (volcanic); the Himalayas are continent–continent (folded, few volcanoes).

Projectile Motion

KEY FACTS

- A projectile moves under gravity alone after launch, along a **parabolic** path.
- With no air resistance, horizontal velocity is constant and vertical velocity changes by 9.8 m/s every second.
- At the peak, vertical velocity is 0 but the acceleration is still *g* downward.
- On level ground, maximum range is at 45°. Complementary angles (30° and 60°) give the same range. Higher angles give a higher peak and longer flight.
- A dropped ball and a ball thrown horizontally from the same height land at the same time.

QUICK CARDS

- Q: Path of a projectile → A: *Parabola*^E
 Q: Only force on an ideal projectile → A: *Gravity*^E
 Q: Launch angle for maximum range → A: 45°^E
 Q: Angle with the same range as 30° → A: 60°^A
 Q: Horizontal velocity of an ideal projectile → A: *Constant*^A
 Q: Vertical velocity at the peak → A: *Zero*^A
 Q: Angle giving the greatest height → A: 90°^A
 Q: Faster release at the same angle: range → A: *Increases*^E
 Q: Acceleration at the peak → A: *g, downward*^D
 Q: Dropped vs. horizontally thrown ball from the same height → A: *Land at the same time*^D

Common confusions

Horizontal vs. **vertical motion**: horizontal is constant velocity; vertical is constant acceleration (*g*). They are independent.

Range vs. **maximum height**: 45° maximizes range; 90° maximizes height.

Momentum and Collisions

KEY FACTS

- Momentum $p = mv$, in kg m/s. It is a vector. More momentum makes an object harder to stop.
- **Impulse** = $F\Delta t = \Delta p$. Stretching the impact time lowers the force (seatbelts, airbags, crumple zones, helmets, bending your knees).
- Total momentum before a collision equals total momentum after (closed system).
- In an **elastic** collision, KE is conserved (bouncing). In an **inelastic** one, KE is lost. In a perfectly inelastic one, the objects stick. Momentum is conserved in all of them.

QUICK CARDS

- Q: Formula for momentum → A: $p = mv$ ^E
 Q: SI unit of momentum → A: kg m/s^E
 Q: Momentum of an object at rest → A: *Zero*^E
 Q: 50 kg runner at 4 m/s → A: 200 kg m/s^A
 Q: Force multiplied by time → A: *Impulse*^E
 Q: Impulse equals the change in → A: *Momentum*^A
 Q: Airbags reduce force by increasing → A: *Impact time*^A
 Q: Car section designed to crush and absorb a crash → A: *Crumple zone*^E
 Q: Loaded truck or bicycle at the same speed: harder to stop? → A: *Truck*^E
 Q: Why bend your knees when landing → A: *Longer stopping time, less force*^A
 Q: Collision in which KE is conserved → A: *Elastic*^E
 Q: Objects stick together after colliding → A: *Perfectly inelastic*^A
 Q: Is momentum conserved in inelastic collisions? → A: *Yes*^D
 Q: Why a gun recoils → A: *Conservation of momentum*^A
 Q: 2 kg cart at 3 m/s sticks to a resting 1 kg cart: final speed → A: 2 m/s^D

Common confusions

Momentum vs. **inertia**: inertia depends on mass only; momentum needs motion (mv).

Elastic vs. **inelastic**: both conserve momentum; only elastic collisions also conserve kinetic energy.

Impulse vs. **force**: the same impulse spread over a longer time means a smaller force.

Electricity: Generation, Transmission, Distribution**KEY FACTS**

- Generators turn mechanical energy into electrical energy by **electromagnetic induction** (Michael Faraday).
- The chain: power plant → step-up transformer (high voltage means less loss) → transmission lines (NGCP) → substation step-down → distribution utility (Meralco or an electric cooperative) → meter → home.
- Bills are in kilowatt-hours (kWh). The PH grid runs on AC at 60 Hz.
- Sources: renewable (geothermal, hydro, solar, wind, biomass) and non-renewable (coal, natural gas from Malampaya off Palawan, oil).

QUICK CARDS

- Q:** Principle behind generators → **A:** *Electromagnetic induction*^E
- Q:** Discovered electromagnetic induction → **A:** *Michael Faraday*^A
- Q:** Device that raises voltage for transmission → **A:** *Step-up transformer*^A
- Q:** Why power is transmitted at high voltage → **A:** *Less energy lost as heat*^D
- Q:** Operator of the PH transmission grid → **A:** *NGCP*^D
- Q:** Distribution utility of Metro Manila → **A:** *Meralco*^E
- Q:** Energy unit on electric bills → **A:** *Kilowatt-hour*^E
- Q:** 1 kW appliance used for 3 h → **A:** 3 kWh^A
- Q:** Frequency of PH household AC → **A:** 60 Hz^D
- Q:** Natural-gas field off Palawan → **A:** *Malampaya*^A
- Q:** Electricity from Earth's internal heat → **A:** *Geothermal*^E

Common confusions

kW vs. **kWh**: kW is power (rate); kWh is energy (what you pay for).

Transmission vs. **distribution**: transmission is high-voltage, long-distance (NGCP); distribution is low-voltage, local (Meralco/co-ops).

AC vs. **DC**: AC reverses direction (wall outlets); DC flows one way (batteries).

4.4 Appendix: K-12 MELC-Only Grade 10 Topics

These topics are in the older Grade 10 MELCs but not in the MATATAG BOW. Study them only if your school or the contest still uses the K-12 MELCs.

Nervous System**KEY FACTS**

- A neuron has dendrites (receive), a cell body, and an axon (sends). The myelin sheath speeds the signal. Neurons meet at synapses, where neurotransmitters cross.
- The CNS is the brain and spinal cord; the PNS is everything else. The autonomic system has sympathetic (“fight or flight”) and parasympathetic (“rest and digest”) branches.
- The cerebrum handles thought and voluntary action, the cerebellum balance and coordination, and the medulla oblongata breathing and heartbeat.

- Reflex arc: receptor → sensory neuron → interneuron → motor neuron → effector.

QUICK CARDS

- Q:** Branch of a neuron that receives signals → **A:** *Dendrite*^E
- Q:** Fatty layer that speeds nerve impulses → **A:** *Myelin sheath*^A
- Q:** Gap between two neurons → **A:** *Synapse*^A
- Q:** Brain part for balance → **A:** *Cerebellum*^E
- Q:** Brain part controlling heartbeat and breathing → **A:** *Medulla oblongata*^A
- Q:** “Fight or flight” division → **A:** *Sympathetic*^D

Endocrine and Reproductive Systems**KEY FACTS**

- The pituitary is the “master gland.” The thyroid makes thyroxine (metabolism; needs iodine). The adrenal glands make adrenaline.
- Ovaries make estrogen and progesterone; testes make testosterone.
- The menstrual cycle is about 28 days. FSH and LH come from the pituitary, and an LH surge triggers ovulation (around day 14). Fertilization usually happens in the fallopian tube.

QUICK CARDS

- Q:** Master gland → **A:** *Pituitary*^E
- Q:** Hormone that triggers ovulation → **A:** *LH*^D
- Q:** Usual site of fertilization → **A:** *Fallopian tube*^A
- Q:** Mineral needed to make thyroxine → **A:** *Iodine*^A
- Q:** Hormone that maintains the uterine lining → **A:** *Progesterone*^D

Protein Synthesis (DNA → RNA → Protein)**KEY FACTS**

- The **central dogma** (Crick): DNA → RNA → protein.
- Transcription** copies DNA into mRNA in the nucleus. **Translation** builds the protein at the ribosome, with tRNA bringing amino acids.
- A codon is 3 mRNA bases. The start codon is AUG (methionine). Stop codons are UAA, UAG, and UGA.

QUICK CARDS

- Q:** DNA → mRNA → **A:** *Transcription*^E
- Q:** mRNA → protein → **A:** *Translation*^E
- Q:** RNA that carries amino acids → **A:** *tRNA*^A
- Q:** Three-base unit on mRNA → **A:** *Codon*^A
- Q:** Start codon → **A:** *AUG*^D

Gas Laws and Kinetic Molecular Theory**KEY FACTS**

- Boyle:** $P \propto 1/V$ (constant T). **Charles:** $V \propto T$ (constant P , T in K). **Gay-Lussac:** $P \propto T$ (constant V). **Avogadro:** $V \propto n$.
- Ideal gas law: $PV = nRT$. At STP (273 K, 1 atm), one mole of gas fills 22.4 L. 1 atm = 760 mmHg.
- Kinetic molecular theory: gas particles move randomly, have negligible volume and no attractions, and collide elastically. Average KE is proportional to the kelvin temperature.

QUICK CARDS

- Q:** Pressure up, volume down at constant T → **A:** *Boyle's law*^E
- Q:** Volume vs. temperature at constant P → **A:** *Charles's law*^E
- Q:** Pressure vs. temperature at constant V → **A:** *Gay-Lussac's law*^A
- Q:** Molar volume of a gas at STP → **A:** 22.4 L^A
- Q:** 1 atm in mmHg → **A:** 760^A

Q: Gas at 2L and 1atm compressed to 2atm (constant T) → **A:** 1L^D

Biomolecules

KEY FACTS

- **Carbohydrates** (C, H, O) are the main energy source: glucose, sucrose, starch (plants), glycogen (animals), cellulose. Tests: Benedict's turns brick-red with reducing sugars; iodine turns blue-black with starch.
- **Lipids** are fats, oils, phospholipids, and steroids, built from glycerol and fatty acids. They store long-term energy (grease-spot or Sudan III test).
- **Proteins** are amino acids joined by peptide bonds; enzymes are proteins (Biuret test turns violet).
- **Nucleic acids** are DNA and RNA, made of nucleotides.

QUICK CARDS

Q: Storage carbohydrate in animals → **A:** *Glycogen*^A

Q: Monomer of proteins → **A:** *Amino acid*^E

Q: Bond linking amino acids → **A:** *Peptide bond*^A

Q: Test for protein (turns violet) → **A:** *Biuret test*^D

Q: Benedict's test detects → **A:** *Reducing sugars*^D

Electromagnetism: Motors and Generators

KEY FACTS

- **Ørsted** (1820) showed that an electric current produces a magnetic field.
- A motor turns electrical energy into mechanical energy; a generator does the reverse.
- Fleming's left-hand rule is for motors; the right-hand rule is for generators.

QUICK CARDS

Q: Showed that current creates a magnetic field → **A:** *Hans Christian Ørsted*^A

Q: Device turning electrical into mechanical energy → **A:** *Electric motor*^E

Q: Fleming's rule for motors → **A:** *Left-hand rule*^D

Common confusions

Motor vs. **generator**: a motor turns electricity into motion; a generator turns motion into electricity.

Boyle vs. **Charles**: Boyle's law is P - V (inverse); Charles's law is V - T (direct).

Transcription vs. **translation**: transcription writes DNA into RNA in the nucleus; translation reads RNA into protein at the ribosome.

Cards in this part: 193 (E 71, A 77, D 45)

5 Rapid Fire (Mixed Grades)

Mixed G7–G10 drill. Each cue is a short scenario, like a multiple-choice stem. The bracket shows the source grade. Cover the answers, set a 20-second timer, and say the answer aloud.

5.1 Easy

- Q: [G7] Ice cream melting on the counter → A: *Melting; heat absorbed*^E
- Q: [G7] Smell of adobo reaching the next room → A: *Diffusion (particles moving)*^E
- Q: [G7] Salt dissolved in water: which is the solute? → A: *Salt*^E
- Q: [G7] Calamansi juice turns blue litmus → A: *Red (acidic)*^E
- Q: [G7] Skin cells healing a wound divide by → A: *Mitosis*^E
- Q: [G7] Carabao eating grass is a → A: *Primary consumer*^E
- Q: [G7] Ground shakes most strongly near the → A: *Epicenter*^E
- Q: [G8] Bulb glows when switched on: energy change → A: *Electrical → light + heat*^E
- Q: [G8] Swing at its lowest point has maximum → A: *Kinetic energy*^E
- Q: [G8] Your reflection in a spoon's back is → A: *Upright and smaller (convex)*^E
- Q: [G8] Strong winds and heavy rain from the Pacific → A: *Tropical cyclone (bagyo)*^E
- Q: [G9] Ball rolls to a stop on grass because of → A: *Friction (net force)*^E
- Q: [G9] Phone charger in the wall: series or parallel with other appliances? → A: *Parallel*^E
- Q: [G9] Sunscreen protects against → A: *Ultraviolet*^E
- Q: [G9] Mangrove-lined river mouth in Bohol is an → A: *Estuary*^E
- Q: [G10] Bread rises due to → A: *CO₂ from yeast*^E
- Q: [G10] Sweating after running is an example of → A: *Homeostasis (negative feedback)*^E
- Q: [G10] Seatbelt keeps a passenger from flying forward. Law? → A: *Inertia (Newton's 1st)*^E

5.2 Average

- Q: [G7] Clothes dry faster on a windy day because → A: *Faster evaporation*^A
- Q: [G7] Sugar dissolves faster in hot tea because → A: *Particles move faster*^A
- Q: [G7] Jeepney goes 30 km in 0.5 h: average speed → A: *60 km/h*^A
- Q: [G7] Philippine Fault Zone movement is mostly → A: *Horizontal (strike-slip)*^A
- Q: [G7] Habagat season months, roughly → A: *June–September*^A
- Q: [G8] Two unaffected parents have an affected child: the trait is → A: *Recessive*^A
- Q: [G8] A leaf kept in CO₂-free air cannot carry out → A: *Photosynthesis*^A
- Q: [G8] Pencil looks bent in a glass of water → A: *Refraction*^A
- Q: [G8] Element with 2, 8, 2 electron arrangement → A: *Magnesium*^A

- Q: [G8] Most explosive PH eruption of the 20th century → A: *Pinatubo (1991)*^A
- Q: [G9] Heavier box needs a bigger push for the same acceleration. Law? → A: *Newton's 2nd*^A
- Q: [G9] One Christmas-light bulb fails and the rest go dark → A: *Series wiring*^A
- Q: [G9] Hospital scans of broken bones use → A: *X-rays*^A
- Q: [G9] Fossil found in the bottom layer is → A: *Older*^A
- Q: [G9] Copper wire is a good conductor because of → A: *Metallic bonding (free electrons)*^A
- Q: [G10] Iron gate rusting in a coastal town: reaction type → A: *Combination (oxidation)*^A
- Q: [G10] Antacid relieving acid indigestion → A: *Neutralization*^A
- Q: [G10] Long-necked giraffes outsurvive others in a drought → A: *Natural selection*^A
- Q: [G10] Freshly sliced apple browns slower in the ref because → A: *Low temperature slows reactions*^A

5.3 Difficult

- Q: [G7] Same mass of sand and water in the sun: which heats faster? → A: *Sand*^D
- Q: [G7] Tsunami warning in PH is issued by → A: *PHIVOLCS*^D
- Q: [G7] Shortest daytime in the PH → A: *December solstice*^D
- Q: [G8] Rr × Rr: fraction of offspring that are heterozygous → A: *$\frac{1}{2}$* ^D
- Q: [G8] Stone dropped from a cliff, ignoring air: its mechanical energy → A: *Stays constant*^D
- Q: [G8] Ion count in ⁴⁰Ca²⁺: protons, electrons → A: *20 p, 18 e*^D
- Q: [G8] Why typhoons rarely form right at the equator → A: *Coriolis effect is near zero*^D
- Q: [G9] Rocket pushes exhaust down; exhaust pushes rocket → A: *Up (equal and opposite)*^D
- Q: [G9] Three 6 Ω bulbs in series across 18 V: current → A: *1 A*^D
- Q: [G9] Radio wave with $f = 100$ MHz: λ (use $c = 3 \times 10^8$ m/s) → A: *3 m*^D
- Q: [G9] Sample with 25% of its C-14 left: age → A: *About 11 460 yr (2 half-lives)*^D
- Q: [G9] Bond type in table sugar → A: *Covalent*^D
- Q: [G10] CaCO₃ → CaO + CO₂ is → A: *Decomposition (endothermic)*^D
- Q: [G10] Doubling the mass at the same velocity: momentum → A: *Doubles*^D
- Q: [G10] Kicking a ball at 20° vs. 70°: range → A: *Equal (complementary angles)*^D
- Q: [G10] Insulin-producing bacteria: traditional or modern biotech? → A: *Modern (recombinant DNA)*^D

Cards in this part: 53 (E 18, A 19, D 16)

6 Cheat Sheets

SI Base Units

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Amount of substance	mole	mol
Electric current	ampere	A
Luminous intensity	candela	cd

SI Prefixes

Prefix	Symbol	Factor	Prefix	Symbol / Factor
giga	G	10 ⁹	deci	d 10 ⁻¹
mega	M	10 ⁶	centi	c 10 ⁻²
kilo	k	10 ³	milli	m 10 ⁻³
hecto	h	10 ²	micro	μ 10 ⁻⁶
deka	da	10 ¹	nano	n 10 ⁻⁹

Mnemonic: King Henry Died By Drinking Chocolate Milk (kilo, hecto, deka, base, deci, centi, milli).

One-Step Formulas

Quantity	Relation	Unit
Speed	$v = d/t$	m/s
Acceleration	$a = \Delta v/t$	m/s ²
Force (2nd law)	$F = ma$	N
Weight	$W = mg$	N
Density	$\rho = m/V$	g/cm ³
Work	$W = Fd$	J
Power	$P = W/t$	W
Kinetic energy	$KE = \frac{1}{2}mv^2$	J
Potential energy	$PE = mgh$	J
Momentum	$p = mv$	kg m/s
Impulse	$F\Delta t = \Delta p$	N s
Ohm's law	$V = IR$	V, A, Ω
Wave speed	$v = f\lambda$	m/s
% by mass	$\frac{m_{\text{solute}}}{m_{\text{solution}}} \times 100$	%
Temperature	$K = ^\circ\text{C} + 273$	K

Periodic Table Groups

Group	Name	Val. e ⁻	Examples
1	Alkali metals	1	Li, Na, K
2	Alkaline earth metals	2	Be, Mg, Ca
3–12	Transition metals	–	Fe, Cu, Au
13	Boron group	3	B, Al
14	Carbon group	4	C, Si
15	Nitrogen group	5	N, P
16	Oxygen group	6	O, S
17	Halogens	7	F, Cl
18	Noble gases	8 (He: 2)	He, Ne, Ar

Periodic Trends

Property	Across a period →	Down a group ↓
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electronegativity	increases	decreases
Metallic character	decreases	increases

Most electronegative: F. Most reactive metal group: alkali metals.

First 20 Elements

1 H Hydrogen	11 Na Sodium
2 He Helium	12 Mg Magnesium
3 Li Lithium	13 Al Aluminium
4 Be Beryllium	14 Si Silicon
5 B Boron	15 P Phosphorus
6 C Carbon	16 S Sulfur
7 N Nitrogen	17 Cl Chlorine
8 O Oxygen	18 Ar Argon
9 F Fluorine	19 K Potassium
10 Ne Neon	20 Ca Calcium

Taxonomic Ranks (Humans)

Rank	Humans
Domain	Eukarya
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Primates
Family	Hominidae
Genus	<i>Homo</i>
Species	<i>Homo sapiens</i>

Electromagnetic Spectrum (low → high frequency)

Wave	Common use	Hazard
Radio	AM/FM, TV, comms	minimal
Microwave	ovens, Wi-Fi, satellites	tissue heating
Infrared	remotes, thermal cameras	burns
Visible	sight, photography	eye strain (intense)
Ultraviolet	sterilizing, vitamin D	sunburn, skin cancer
X-ray	bone imaging, security	ionizing (DNA damage)
Gamma	cancer therapy, sterilizing	most ionizing

Wavelength decreases and energy increases going down the table.

Layers of the Earth

Layer	State	Composition	Approx. thickness
Crust	solid	granite/basalt	5 km to 70 km
Mantle	solid/plastic	Fe–Mg silicates	~2900 km
Outer core	liquid	Fe, Ni	~2200 km
Inner core	solid	Fe, Ni	~1220 km (radius)

Layers of the Atmosphere

Layer	Approx. height	Key fact
Troposphere	0–12 km	weather; temperature drops with height
Stratosphere	12–50 km	ozone layer
Mesosphere	50–85 km	meteors burn up; coldest layer
Thermosphere	85–600 km	auroras, ISS orbit
Exosphere	beyond	fades into space

Mnemonic: The Strong Man's Triceps Explode (Tropo, Strato, Meso, Thermo, Exo).

Rock Types

Type	Formed by	Examples
Igneous	cooling magma/lava	granite, basalt, pumice, obsidian
Sedimentary	compacted layers	sandstone, shale, limestone
Metamorphic	heat + pressure	marble (limestone), slate (shale)

Volcano Types

Type	Shape / material	Example
Shield	broad, gentle; fluid basalt	Mauna Loa
Cinder cone	small, steep; pyroclastics	Parícutin
Composite	steep, layered lava + ash	Mayon

Fault Types

Fault	Stress	Movement
Normal	tension	hanging wall moves down
Reverse/thrust	compression	hanging wall moves up
Strike-slip	shear	horizontal, side by side

Plate Boundaries

Boundary	Motion	Features	Example
Divergent	apart	ridges, rift valleys	Mid-Atlantic Ridge
Convergent	together	trenches, arcs, mountains	Himalayas, Andes
Transform	slide past	faults, quakes	San Andreas

PAGASA Tropical Cyclone Categories

Category	Max sustained winds
Tropical Depression	≤ 61 km/h
Tropical Storm	62–88 km/h
Severe Tropical Storm	89–117 km/h
Typhoon	118–184 km/h
Super Typhoon	≥ 185 km/h

Acid–Base Indicators

Indicator	In acid	In base
Litmus	blue → red	red → blue
Phenolphthalein	colourless	pink
Methyl orange	red	yellow
Red cabbage	red/pink	green/yellow
Turmeric	yellow	reddish-brown

Who’s Who (Fast Recall)

Scientist	Known for
Hooke	named the “cell”
Linnaeus	taxonomy, binomial names
Mendel	genetics (peas)
Dalton	solid-sphere atom
Thomson	electron, plum pudding
Rutherford	nucleus (gold foil)
Bohr	energy levels
Chadwick	neutron
Mendeleev	periodic table
Moseley	atomic-number order
Newton	3 laws of motion
Ohm	$V = IR$
Faraday	electromagnetic induction
Wegener	continental drift
Hess	seafloor spreading
Watson & Crick	DNA double helix
Darwin	natural selection
Lavoisier	conservation of mass

7 Sources

Checked October 2026. The *depd.gov.ph* site was not reachable from the build environment, so official-page URLs below are as listed by search results and should be opened once to confirm.

Curriculum in force, SY 2026–2027

- **Phase-in schedule (MATATAG, DO 10 s. 2024):** K, G1, G4, G7 in SY 2024–25; G2, G5, G8 in SY 2025–26; G3, G6, G9 in SY 2026–27; G10 in SY 2027–28.
DepEd FAQs on the MATATAG Curriculum: <https://www.deped.gov.ph/wp-content/uploads/FAQs-ON-THE-MATATAG-CURRICULUM.pdf>
DepEd MATATAG page (curriculum guides): <https://www.deped.gov.ph/matatagcurriculumk147/>
News report on DO 10 s. 2024: <https://newsinfo.inquirer.net/1984864/depd-issues-new-guidelines-for-matatag-curriculum-implementation>
- **Three-term calendar:** DepEd Order No. 9, s. 2026 (school year divided into Terms 1–3; classes from 8 June 2026). Reported in: <https://www.gmanetwork.com/news/topstories/nation/984622/explainer-deped-school-calendar-sy/story/>

Per-grade scope

- **Grade 7 – MATATAG** (third year of use). DepEd, *Science Grade 7 Three-Term Budget of Work for Learning Competencies*, last updated 8 April 2026 (PDF supplied by the user). Parent document: MATATAG Science Curriculum Guide, Grades 3–10 (August 2023), via the DepEd MATATAG page above; mirror: <https://www.slideshare.net/slideshow/final-matatag-science-cg-2023-grades-3-10-pdf/270247552>
- **Grade 8 – MATATAG** (second year of use). DepEd, *Science Grade 8 Three-Term Budget of Work*, last updated 8 April 2026 (PDF supplied by the user). Same parent curriculum guide.
- **Grade 9 – MATATAG** (first year of use). DepEd, *Science Grade 9 Three-Term Budget of Work*, last updated 17 April 2026 (PDF supplied by the user). Teacher-site listing of the same BOW: <https://depdclub.com/grade-9-budget-of-work-bow-term-1-3-matatag-three-term-sy-2026-2027/>
- **Grade 10 – conflicting sources.** Core: DepEd, *Science Grade 10 Three-Term Budget of Work* (MATATAG content), last updated 10 April 2026 (PDF supplied by the user); listing: <https://depdclub.com/grade-10-budget-of-work-bow-term-1-3-matatag-three-term-sy-2026-2027/>. The official schedule above puts Grade 10 MATATAG in SY 2027–28, so the appendix adds K–12 MELC-only topics from the Grade 10 Science MELCs (third-party copy): <https://www.teacherph.com/wp-content/uploads/2022/08/Grade-10-Science-Most-Essential-Learning-Competencies-MELCs.pdf>. Which Grade 10 curriculum your school uses this year. **[VERIFY]**

Philippine-specific fact checks

- PAR boundary coordinates – PAGASA: <https://www.pagasa.dost.gov.ph/learning-tools/philippine-area-of-responsibility>
- Tropical cyclone categories (super typhoon ≥ 185 km/h, since 2022) – <https://www.gmanetwork.com/news/scitech/weather/914610/explainer-what-is-a-super-typhoon/story/>
- Active volcano count (PHIVOLCS, 24) – <https://www.philstar.com/headlines/2020/01/14/1984829/headlines>
- Philippine Fault Zone (~1200 km, left-lateral strike-slip) – https://en.wikipedia.org/wiki/Philippine_fault_system
- IUCN statuses of PH species – check each on <https://www.iucnredlist.org>

Scope of fact-checking

General science content (definitions, laws, classic experiments, and who proposed them) is standard junior-high textbook material and is not cited line by line. Only items that are Philippine-specific, recently changed, or count-based were checked online. Any such item that could not be confirmed is tagged **[VERIFY]**.

Items to verify

Every **[VERIFY]** tag in this document, with its page. These are facts that were uncertain, that changed recently, or where sources disagree. Check each against the source named in *Sources* (or ask your coach) before the contest.

- Reverse/thrust and strike-slip, from plate convergence (expected DepEd answer).....p. 5
- 24 active PH volcanoes.....p. 9
- 24 (PHIVOLCS active list).....p. 10
- Philippine pangolin (IUCN: CR)p. 14
- Conversion to fishponds (main historic cause of PH mangrove loss)p. 15
- Confirm with your school which Grade 10 curriculum applies this year.....p. 17
- about 1.1 °C above pre-industrial levels (IPCC AR6, 2011–2020)p. 19
- About 1.1 °C of global warmingp. 19
- about 7 cm to 8 cm per year (Philippine Sea Plate speed)p. 20
- Which Grade 10 curriculum your school uses this year.p. 26